INDIGENE

CHM 101
PASS MANUAL

Introductory Inorganic and Physical Chemistry

HOW TO USE THIS MANUAL

Read one chapter at a time. Start with the **Alzheimer's Mode** box, it gives you the whole chapter in plain words.

Worked examples show the exact steps examiners expect. Copy the steps, not the numbers.

Exam Hack boxes are shortcuts. **Trap** boxes are the mistakes that cost marks.

End every chapter with the **Quick Recall** box, then drill the CBT for that chapter.

Everything here comes from the course textbook. Nothing is added from outside it.

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01 Physical Quantities and Error Treatments

CHAPTER 1 · READ WITH THE BOOK: PDF pages 5-11

ALZHEIMER'S MODE

Think of **measuring** as asking: "how many of this ruler fit into that thing?" The **quantity** is what you measure (length). The **unit** is the ruler you use (metre). The number you write down only makes sense with both.

Then comes honesty. No measurement is perfect, so a scientist always says **how sure** they are. This chapter is about writing numbers honestly (significant figures) and naming the two ways a measurement can go wrong (systematic and random error).

1.1 Physical quantity, unit and measurement

- A **physical quantity** is an attribute of a physical body, phenomenon or process that can be measured. Length, width and height of an object are examples.
- The **value** of a physical quantity is its size expressed as a number of units.
- A **unit** is an established standard against which a quantity is compared. Comparing a quantity to a unit is called **measurement**.
- A set of fundamental units that is enough to measure the properties of the field under study is called a **system of units**.

1.2 The International System of Units (SI)

SI comes from the French **Système International d'Unités**. It is the modern metric system and is the main system used in science, technology and international commerce. It is built on **seven base units** for seven base quantities, assumed to be independent of each other.

Base quantity	Symbol	SI base unit
Length	l	metre (m)
Mass	m	kilogram (kg)
Time	t	second (s)
Electric current	I	ampere (A)
Thermodynamic temperature	T	kelvin (K)
Luminous intensity	Iv	candela (cd)
Amount of substance	n	mole (mol)

EXAM HACK

Seven base units in order: Length, Mass, Time, Electric current, Temperature, Luminous intensity, Amount of substance. The last one is the **mole**, the unit this whole course keeps coming back to.

1.3 Base and derived quantities

- **Base quantities** are chosen quantities with **fundamental units**.
- **Derived quantities** are built from the base ones using equations. Their units are **derived units**.
- Example: if displacement and time are base quantities, then **velocity = displacement ÷ time** is derived.
- Acceleration has the derived unit **m/s²**.

TRAP · examiners love this

Do not call velocity a base quantity. It is derived. Mass, length and time are base.

1.4 Significant figures

The number of meaningful digits you write tells the reader how accurate the measurement is. A length of 0.528 m implies an uncertainty of about 0.001 m. Writing 0.5 or 0.52819667 would claim the wrong accuracy. Report only as many figures as your error allows.

- Significant figures have **nothing to do with decimal places**. 0.528 m and 52.8 cm both have **three** significant figures.
- Only **one uncertain digit** is reported. If the error is 0.02 m, write **0.53 ± 0.02 m**, not 0.528 ± 0.02 m.

ZERO RULE
A zero is significant if a non-zero digit is somewhere to its LEFT.

Number	Sig. figs	Why
5.00	3	Zeros have 5 on their left
0.0005	1	Zeros on the left only: not significant
1.0005	5	Zeros are between non-zero digits
6000	not well defined	Write 6×10^3 (1 s.f.) or 6.00×10^3 (3 s.f.)

EXAM HACK

Cover the number from the **left**. Skip every zero until you hit the first non-zero digit. Count everything from there to the end.

TRAP · examiners love this

6000 is a favourite. The book says it is not well defined. The fix is scientific notation.

1.5 Precision and accuracy

Word	Meaning (from the book)
Precision	How close **repeated measurements** of the same quantity are to each other (reproducibility)
Accuracy	How close a measurement, or the mean of several, is to the **true or accepted value**

Book example: one scientist got 22.60, 22.75 and 22.98 per cent. A second got 22.63, 22.62 and 22.62 per cent. The second set varies much less, so it is **more precise**.

ALZHEIMER'S MODE

Archery target. Arrows bunched together = **precise**. Arrows near the bull's-eye = **accurate**. You can be precise and still miss the bull's-eye.

TRAP · examiners love this

A series can be **very precise and highly inaccurate**. But it is meaningless to talk about accuracy if the precision is poor. Do not write "precise means accurate".

1.6 Error

Error is always present. A scientist does not ask "is there error?" but "**how much** error is there?" He tries to recognise, reduce and evaluate it, but never removes it completely.

Absolute and relative error

Type	Meaning	Units
Absolute error	The uncertainty in the quantity. Example: in 0.428 ± 0.002 m, the 0.002 m is the absolute error	Same as the quantity
Relative (fractional) error	Absolute error ÷ the quantity. Multiply by 100 to give a percentage	None (dimensionless)

Relative error = absolute error ÷ quantity
Percentage error = relative error × 100

WORKED EXAMPLE · relative error of 0.428 ± 0.002 m

Relative error = 0.002 ÷ 0.428 = 0.00467

Percentage error = 0.00467 × 100 = **0.47 %**

Relative error is usually more significant than absolute. The book compares a 1 mm error in a skate wheel diameter (serious) with 1 mm in a truck tyre (minor).

1.7 Systematic (determinate) error

- Errors that are **one-sided** and can be **traced to a specific source**, either in the plan of the experiment or in the apparatus.
- They make results **consistently too high or too low**.
- They can make results **highly inaccurate without affecting precision**.
- Can often be reduced by a modified plan, or corrected with a correction factor.
- The error either stays **constant**, **varies systematically** with a condition, or **persists as a random error**.
- A constant error points to **instrument calibration**.
- Fix: vary experimental factors in repeats, and in critical work analyse duplicate samples by **entirely different methods**.

1.8 Random (indeterminate) error

- The cause may or may not be known.
- Comes from **personal judgement**: reading dials or meters, deciding when liquid reaches a mark.
- Comes from the method itself: impurity of a supposedly pure material, stirring, speed of mixing reagents.
- Comes from random changes in room temperature and other surroundings.
- An apparently random error can be a systematic one linked to an uncontrolled condition. The book's example: changes in **atmospheric humidity** because a chemical absorbs water.

	Systematic	Random
Direction	One-sided (always high or always low)	Scatters both ways
Traceable?	Yes, to a specific source	Cause may be unknown
Hurts	Accuracy	Precision (scatter)
Cure	Calibrate, change plan, different method	Repeat and average

EXAM HACK

Systematic = **S**ame direction. **R**andom = **R**andom scatter.

ALL THE HACKS FOR THIS CHAPTER

SI base units: Length, Mass, Time, Electric current, Temperature, Luminous intensity, Amount of substance = **7**. Think "**L**ittle **M**en **T**ake **E**very **T**ray **L**eft **A**round" (L, M, T, E, T, L, A).

Zero rule: start counting from the **first non-zero digit**. Everything after it counts, including trailing zeros after a decimal point.

6000? Never leave it bare. Write 6×10^3 or 6.00×10^3.

Precise vs accurate: precise = **P**acked together. Accurate = hits the **A**ctual value.

Systematic = Same direction (always high or always low). **Random = scatter** both ways.

Percentage error = absolute error ÷ value × 100. No units on relative error.

QUICK RECALL · read this the night before

SI = Système International d'Unités, built on **7 base units**.

Base units: m, kg, s, A, K, cd, mol. Derived example: acceleration m/s^2.

Zero is significant only if a non-zero digit is to its left. 5.00 has 3, 0.0005 has 1, 1.0005 has 5.

Report only **one** uncertain digit.

Precision = reproducibility. Accuracy = closeness to true value.

Absolute error has units. Relative error has none. Percentage error = relative × 100.

Systematic error: one-sided, traceable, hurts accuracy. Random error: scatter, hurts precision.

02 Structure of the Atom

CHAPTER 2 · READ WITH THE BOOK: PDF pages 12-35

ALZHEIMER'S MODE

This chapter is a **detective story**. Each scientist adds one clue about what an atom is made of: Democritus says "tiny and indivisible", Dalton says "solid balls", Thomson finds the electron, Rutherford finds the nucleus, Bohr puts electrons on fixed shells, and the wave model says "stop picturing planets".

The second half is a **filing system**: electrons go into shells, subshells and orbitals by fixed rules, and the periodic table is just that filing system drawn as a chart. Learn the rules once and you can write any configuration and predict every trend.

2.1 Early ideas: Democritus and Dalton

- Early Greek philosophers used **reason and logic**, not experiment.
- **Democritus** believed the universe is made of matter, and matter is made of **atoms**: tiny, indivisible, solid objects.
- **John Dalton** used both logic and experiment and proposed the **first scientific theory** linking chemical changes to the atom.

Dalton's Atomic Theory

#	Statement
1	Matter is made of tiny particles (atoms)
2	Atoms are indivisible. In a reaction they are **rearranged**, not broken apart, created or destroyed
3	All atoms of an element are **identical in mass and other properties**
4	Atoms of different elements **differ in mass and other properties**
5	Atoms of one element combine with atoms of another to form compounds, always in the **same relative numbers**

TRAP · examiners love this

Dalton said atoms are indivisible. That was later shown **not entirely true** by Thomson in 1897, because atoms contain electrons.

2.2 Faraday and the cathode rays

- **Michael Faraday** gave the **first evidence** that electric charges are not continuous but exist as **discrete particles**.
- From electrolysis, the electricity needed to deposit **one mole** of a univalent element is always the same. So a fixed amount of electricity belongs to one univalent atom.
- For divalent or trivalent atoms, you need only **whole multiples** of that amount.
- Cathode ray work in the later 19th century was done by scientists such as **Crookes** and **Goldstein**. High voltage across gas at low pressure makes a stream of particles go **from cathode to anode**: the cathode rays. At low pressure a glow is seen.

2.3 J. J. Thomson (1897)

- Used a **cathode ray tube**: glass tube, cathode (negative plate) at one end, anode (positive plate) at the other.
- Steps: let the beam make a spot on the screen in a straight line, **deflect it with a magnetic field**, then **balance it back with an electric field**.
- From this he found the **charge-to-mass ratio (e/m)** of the electron.
- The e/m value was **the same whatever the cathode material**, so the particle is a universal part of matter. It was named the **electron**.
- He modelled the atom as a **sphere of positive charge with electrons embedded in it** (the plum-pudding idea). Mass was spread uniformly with the positive charge.

2.4 Robert Millikan: oil drop experiment

- Oil droplets were sprayed between two charged plates. Gravity pulls a drop down with force **mg**.

- X-rays made the oil negatively charged. The electric field gave an **upward force Ee** that balanced gravity, so **Ee = mg** and the charge could be calculated.
- The charges on different drops were always **whole-number multiples of 1.6 × 10^{-19} C**, the charge on one electron.
- This gave the **final evidence that electrons are discrete particles**.
- Combined with Thomson's e/m, the mass of the electron is about **9.1 × 10^{-28} g**.

> **EXAM HACK**
>
> **T**homson = **T**ube, found e/m. **M**illikan = **M**ass and charge from oil **M**icro-drops. Thomson gave the ratio; Millikan gave the charge itself.

2.5 Rutherford's nuclear model

Rutherford tried to **disprove Thomson's model**. He fired **alpha particles** at thin **gold foil**.

Observation	Meaning
Most alpha particles passed straight through	Most of the atom is **empty space**
A few were slightly deflected	Something small and positive is in the way
Occasionally one bounced back 90° or more	A tiny, dense, positive **nucleus** at the centre

> **ALZHEIMER'S MODE**
>
> Rutherford said it was like firing a high-powered rifle at **tissue paper** and seeing a bullet bounce back. The positive charge is **not spread like butter**. It is squeezed into a tiny centre.

2.6 Moseley and the atomic number

- **H. G. J. Moseley** used different elements as the **cathode** under high voltage. X-rays passed through a slit, were diffracted, and fell on a **photographic plate**, giving an X-ray spectrum that is **characteristic of the target element**.
- He found that the frequency of the X-rays depends on the atomic number, not the atomic mass, and showed that the **atomic number gives the order of the elements in the periodic table**.
- Moseley's relation: $\sqrt{\nu} = a(Z - b)$ where ν is the frequency, Z the atomic number, a and b are constants.

2.7 The three sub-atomic particles

Particle	Charge	Mass (amu)	Where
Proton	positive	about 1	Nucleus
Neutron	none	about 1	Nucleus
Electron	negative	about 1/1836 (almost zero)	Outside nucleus

An **atomic mass unit (a.m.u.)** is defined as **one-twelfth of the mass of a carbon-12 atom**. In this unit the proton and the neutron are almost exactly **1.0**.

2.8 Atomic spectra

- Atoms of different elements heated in a flame give **different colours**. Passing the light through a prism gives a **line spectrum**: a series of coloured lines.
- The pattern is **characteristic of each element**, so it can **identify the element**.
- The **simplest line spectrum is that of hydrogen**.

Series	Region
Lyman	Ultra-violet
Balmer	Partly visible
Paschen	Infra-red

Rydberg: $1/\lambda = R\,(1/n_1^2 - 1/n_2^2)$
n_1 and n_2 are whole numbers, $n_2 > n_1$

- **Planck** studied radiation from hot bodies and found energy is not emitted continuously but in **packets called quanta**.
- **Rydberg's constant** $R = 1.097 \times 10^7\ m^{-1}$ (standard value, used below).

EXAM HACK

Spectrum series: **L**yman = n_1 = **1** = **U**ltra-violet. **B**almer = 2 = **B**right (visible). **P**aschen = 3 = **P**remium heat (infra-red). Order by energy: Lyman, Balmer, Paschen.

WORKED EXAMPLE · hydrogen lines from the Rydberg equation (book exercises)

Balmer, first line ($n_2 = 3 \rightarrow n_1 = 2$): $1/\lambda = 1.097\times10^7\,(1/4 - 1/9) = 1.097\times10^7 \times 5/36 \rightarrow \lambda =$ **6.56×10^{-7} m (656 nm)**.

Balmer, second line ($n_2 = 4 \rightarrow n_1 = 2$): $1/\lambda = 1.097\times10^7\,(1/4 - 1/16) = 1.097\times10^7 \times 3/16 \rightarrow \lambda =$ **4.86×10^{-7} m (486 nm)**.

Lyman, first line ($n_2 = 2 \rightarrow n_1 = 1$): $1/\lambda = 1.097\times10^7\,(1 - 1/4) \rightarrow \lambda =$ **1.22×10^{-7} m (122 nm)**.

Lyman, second line ($n_2 = 3 \rightarrow n_1 = 1$): $1/\lambda = 1.097\times10^7 \times 8/9 \rightarrow \lambda =$ **1.03×10^{-7} m (103 nm)**.

Lyman lines are in the ultra-violet because the wavelengths are shorter than visible light.

2.9 Bohr's model

Bohr's model was put forward to explain the **line spectra** of atoms, the contradiction between classical physics and electrons in atoms, and the **periodic law**.

Bohr's postulates

- 1. Electrons move in **orbits** around the central nucleus.
- 2. Only certain orbits are allowed: those where the **angular momentum** is a whole multiple of h/2π, **mvr = nh/2π** (n = 1, 2, 3,

…). This is the **quantum condition**.

- 3. The energy of an electron in a given orbit is **fixed**. While it stays there it **neither absorbs nor radiates energy**.

Energy of orbit n: $E_n = -Rch / n^2$
Jump energy: $\Delta E = E_{n2} - E_{n1} = h\nu$
Wave link: $c = \lambda\nu$

- When an electron moves to another orbit it absorbs or radiates energy equal to the **difference between the two orbits**.
- Types of energy: **electronic energy** is due to the motion of electrons about the nucleus. Translational, rotational and vibrational energies are limited by quantum theory.
- Maximum electrons in a shell = **$2n^2$**. Shells K, L, M, N (n = 1, 2, 3, 4) hold **2, 8, 18, 32**.

Shell	n	Max electrons = $2n^2$
K	1	2
L	2	8
M	3	18
N	4	32

WORKED EXAMPLE · energy of jumps add up (book example)

Show $\Delta E(3\to1) = \Delta E(3\to2) + \Delta E(2\to1)$.

$\Delta E(3\to2) = K(1/2^2 - 1/3^2) = K(1/4 - 1/9)$

$\Delta E(2\to1) = K(1/1^2 - 1/2^2) = K(1 - 1/4)$

Add: $K(1/4 - 1/9 + 1 - 1/4) = K(1 - 1/9) = \Delta E(3\to1)$. ✔

The 1/4 cancels. Energy is a state function, so the route does not matter.

2.10 Wave mechanical model

- Electrons do **not** move in circular orbits like planets round the sun.
- Electrons move randomly, but there is a **higher chance of finding them close to the nucleus**. So they behave as **waves**.

- A wave is described by wavelength λ, frequency ν and speed c, linked by **$\lambda\nu = c$**.
- Planck's constant **$h = 6.63 \times 10^{-34}$ J s**. Speed of light **$c = 3.00 \times 10^{8}$ m s^{-1}**.

WORKED EXAMPLE · frequency from wavelength (book exercise a)

Wavelength 631 nm = 631×10^{-9} m.

$\nu = c / \lambda = 3.00 \times 10^{8} \div 6.31 \times 10^{-7}$

$\nu =$ **4.75×10^{14} Hz**

Always convert nm to m first.

TRAP · examiners love this

Nanometre to metre: multiply by 10^{-9}. Forgetting this is the most common slip in frequency questions.

2.11 Energy levels and subshells

- Atoms have **discrete energy levels**. The lowest level (E1) is the **ground state**.
- Each level has subshells s, p, d, f. They hold at most **s = 2, p = 6, d = 10, f = 14** electrons.
- Some subshells overlap in energy (for example 4s fills before 3d).

2.12 The four quantum numbers

Quantum number	Symbol	Tells you
Principal	n	Size and energy of the orbit
Azimuthal	l	Shape of the orbital
Magnetic	m	Orientation of the orbital, from −l to +l
Spin	s	Spinning direction, +½ or −½

- **s** sublevels are **spherical**. **p** sublevels have **two lobes**. The label x, y or z on a p orbital shows the **axis along which its lobes point**.

ALZHEIMER'S MODE

Think of an address. **n** = the building (how far out), **l** = the shape of the room, **m** = which way the room faces, **spin** = which way the person in the room is turning.

2.13 Rules for filling electrons

Rule	What it says
Aufbau principle	Add electrons to orbitals in order of **increasing energy**, lowest first
Pauli exclusion principle	No two electrons in an atom can have the **same set of four quantum numbers**
Hund's rule	In a set of equal-energy orbitals, electrons enter **singly first**, each orbital half-filled before any is completed
Heisenberg's uncertainty principle	It is impossible to know accurately both the **position and momentum** of a particle at the same time

EXAM HACK

Aufbau = **A**scend from the bottom. **P**auli = **P**airs must spin **opposite**. **H**und = **H**ouse buses: one person per seat first, then pair up.

2.14 Writing electron configurations

Element	Z	Configuration
H	1	$1s^1$
He	2	$1s^2$
Li	3	$1s^2\ 2s^1$
C	6	$1s^2\ 2s^2\ 2p^2$ ($2p_x^1\ 2p_y^1$)
Ar	18	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$
K	19	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

- **Box notation:** each box is an orbital, an up arrow is one electron, an up and down pair is two paired electrons.

- **Noble gas notation:** use the noble gas before the atom to stand for the filled levels. F (9) = [He] $2s^2$ $2p^5$. Mg (12) = [Ne] $3s^2$.
- For ions: **positive ions lose electrons** from the outer shell, **negative ions gain** electrons into the outer shell.

WORKED EXAMPLE · configurations of ions (book exercises)

K (19) = $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^1$. **K^+** loses the 4s electron: $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ (same as argon).

Ca (20) = …$3p^6$ $4s^2$. **Ca^{2+}** loses both 4s electrons: $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$.

Cl (17) = $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^5$. **Cl^-** gains one: …$3s^2$ $3p^6$.

O (8) = $1s^2$ $2s^2$ $2p^4$. **O^{2-}** gains two: $1s^2$ $2s^2$ $2p^6$ (same as neon).

Pattern: ions of nearby elements copy the **nearest noble gas**.

2.15 Valence electrons

The **valence shell** is the outermost shell. Electrons in it are the **valence electrons**. They decide the **combining power** of the atom and how it takes part in bonding.

Atom	Valence electrons
H ($1s^1$)	1
He ($1s^2$)	2
Na (…$3s^1$)	1
Al (…$3s^2$ $3p^1$)	3
Ca (…$4s^2$)	2

TRAP · examiners love this

Helium has **two** valence electrons ($1s^2$). It is stable because the first shell holds only 2. Do not write 1.

2.16 Development of the periodic table

Year	Scientist	Contribution
1824	Johann Döbereiner	**Triads**: groups of three with similar properties. The atomic weight of the middle one is almost exactly the **average** of the other two
1864	John Newlands	**Law of octaves**: arranged by increasing atomic weight, the **8th element** repeats the 1st, like notes in a music scale. Heavier elements did not fit
1869	Dmitri Mendeleev (Russian)	First table much like today's. **63 elements** were known. Ordered by increasing atomic mass, similar elements underneath each other
Later	Bohr	Chemical properties depend on the number of **outermost electrons**, which he used to set up the table
Later	Moseley	The table is ordered by **atomic number**

Döbereiner triads (book)
Chlorine, bromine, iodine
Sulphur, selenium, tellurium
Calcium, strontium, barium
Lithium, sodium, potassium

> **EXAM HACK**
>
> Order of the periodic table story: **D**öbereiner (1824) → **N**ewlands (1864) → **M**endeleev (1869). Remember **"Dear Nigerian Mothers"**.

2.17 Families of the periodic table

Group	Name	Note
1A	Alkali metals	One valence electron, very reactive, soft and easily cut
2A	Alkaline earth metals (Beryllium family)	Two valence electrons
3	Boron family	
4	Carbon family	
5	Nitrogen family	
6	Oxygen family	The book lists this as nitrogen family by mistake
7	Halogen family	
8	Noble (inert) gases	Full outer shell

2.18 Metals, non-metals, metalloids

- A diagonal line on the right of the table separates **metals (left)** from **non-metals (right)**. Hydrogen is a non-metal.
- Elements **on** the line are **metalloids**. **Aluminium** touches the line but is a **metal**.
- Metallic character **increases down a group** and **right to left across a period**, so **francium (Fr)** is the most metallic.
- Metals are shiny, dense, high melting point, **ductile** (drawn into wires), **malleable** (hammered into sheets), good conductors of heat and electricity. All are solids at room temperature **except mercury**.
- Metals **lose electrons** to form **positive ions**. Examples: silver, gold, zinc.

2.19 Periodic trends

Property	Down a group	Across a period (left to right)	Reason
Atomic size	Increases	Decreases	More shells down; stronger pull across
Ionisation energy	Decreases	Increases	Smaller atom, electrons nearer the nucleus
Electron affinity	Generally decreases	Increases	Smaller atom attracts an added electron more
Electronegativity	Decreases	Increases	Greater nuclear charge, smaller size

- **Atomic size:** half the distance between the centres of two bonded atoms; set by the outermost occupied orbital.
- **Ionisation energy:** energy to remove the **most loosely held electron** from a gaseous atom, e.g. Na + energy → $Na^+ + e^-$. Metals have **low** ionisation energies. Valence electrons are removed more easily than inner ones.
- **Ionic radius:** for similarly charged ions, size **increases from top to bottom**. For ions across a period, radius **decreases** with increasing nuclear charge.
- **Electron affinity:** energy change when an electron is **added to an atom in its ground state**. The smaller the atom, the greater its pull on an added electron.
- **Electronegativity:** the relative ability of an atom to **attract shared electrons**. Used to predict whether a bond is mainly ionic or covalent. The most electronegative elements are in the **upper right corner** of the table (fluorine).

TRAP · examiners love this

Several trend lines in the textbook are worded the wrong way round (for example "atomic size decreases from right to left" or "electronegativity decreases from bottom to top"). In the exam, use the correct rule: **size grows down and shrinks to the right; ionisation energy, electron affinity and electronegativity grow up and to the right**.

Ionisation energy, **electronegativity** and electron affinity all rise toward the **top right**. Atomic size does the opposite.

EXAM HACK

Picture an arrow pointing to **fluorine** (top right). Ionisation energy, electronegativity, electron affinity all grow **toward F**. Atomic size grows **away from F** (toward francium, bottom left).

ALL THE HACKS FOR THIS CHAPTER

Thomson = Tube + e/m. Millikan = oil drop + charge 1.6 × 10^{-19} C. Rutherford = gold foil + nucleus. T-M-R, in that order.

Shell capacity $2n^2$: 2, 8, 18, 32.

Subshell capacity: s 2, p 6, d 10, f 14 (double the odd numbers 1, 3, 5, 7).

Quantum numbers: n = size, l = shape, m = orientation, s = spin.

Spectral series: Lyman UV (n = 1), **B**almer visible (n = 2), **P**aschen IR (n = 3).

Ions: write the atom's configuration first, then **remove** electrons (cation) or **add** them (anion) at the outer shell.

Trends: everything except atomic size grows toward **fluorine** (top right).

QUICK RECALL · read this the night before

Greek idea: Democritus. Dalton: indivisible atoms, same relative numbers in compounds.

Faraday: charges are discrete. Thomson 1897: e/m, plum pudding. Millikan: charge 1.6×10^{-19} C. Rutherford: nucleus (gold foil, alpha). Moseley: atomic number.

a.m.u. = 1/12 of carbon-12. Proton ≈ neutron ≈ 1 amu.

Line spectrum identifies an element. Lyman UV, Balmer partly visible, Paschen IR. $1/\lambda = R(1/n_1^2 - 1/n_2^2)$.

Bohr: fixed orbits, $mvr = nh/2\pi$, $E = -Rch/n^2$, shells hold $2n^2$ (2, 8, 18, 32).

Subshell max: s 2, p 6, d 10, f 14. Quantum numbers: n, l, m, spin.

Rules: Aufbau, Pauli, Hund, Heisenberg.

Döbereiner 1824 triads, Newlands 1864 octaves, Mendeleev 1869 (63 elements).

Group 1A alkali, 2A alkaline earth, 7 halogens, 8 noble gases. Fr most metallic. F most electronegative.

03 States of Matter

CHAPTER 3 · READ WITH THE BOOK: PDF pages 36-47

ALZHEIMER'S MODE

Everything around you is **matter**. The only question is how tightly its particles are packed and how freely they move. This chapter spends most of its time on **gases**, because a gas is the easiest state to describe with equations.

Remember the gas laws as **three friends of volume**: pressure squeezes it (Boyle), heat expands it (Charles), and more gas fills more space (Avogadro). Put them together and you get **PV = nRT**. Everything else is a variation on that.

3.1 What is matter?

- **Matter** is anything that has **mass and occupies space**. Each pure kind of matter is a **substance**.
- A **state of matter** is one of the distinct forms matter takes. Four are seen in everyday life: **solid, liquid, gas and plasma**.
- **Plasma** has neutral atoms plus a significant number of **ions and electrons** that move freely. It is the most common form of visible matter in the universe.

State	Volume	Shape	Particles
Solid	Fixed	Fixed	Close together, fixed in place
Liquid	Fixed	Variable (takes the container's shape)	Close together but move freely
Gas	Takes container	Takes container	Loosely packed, move freely
Plasma	Variable	Variable	Neutral atoms plus ions and electrons

3.2 Measurable properties of gases

Property	Unit and notes
Mass	gram or kg
Volume	Equals the container volume. Units: L, mL, cm^3, m^3, dm^3. **1 L = 1000 mL = 1000 cm^3 = 1 dm^3**
Pressure	Gas pressure: **manometer**. Atmospheric pressure: **barometer**
Temperature	Celsius or Kelvin. **SI unit is kelvin**. **T(K) = t(°C) + 273**

1 atm = 760 mm Hg = 760 torr = 101.325 kPa = 101 325 Pa
1 atm = 1.01325 bar | 1 bar = 10^5 Pa

- **STP** (the book's definition): 273.15 K (0 °C) and 1 bar. At STP the molar volume of an ideal gas is about 22.7 L/mol. For school calculations the book also uses **22.4 L** for 1 mole at 0 °C and 1 atm.
- Basic gas properties: expansion, low density, compression, diffusion.

TRAP · examiners love this

Temperature in gas laws is **always in kelvin**. Adding 273 is the step students forget, and it ruins the whole answer.

3.3 Kinetic theory of gases

#	Postulate
1	A gas is made of a very large number of tiny, discrete particles (molecules)
2	The volume of the molecules is **negligible** compared with the volume of the gas
3	Molecules move in **continuous, rapid, random motion** and collide with each other and the walls
4	Collisions are **perfectly elastic**. No kinetic energy is lost
5	There are **no attractive forces** between the molecules
6	Pressure is caused by molecules **bombarding the walls** of the container
7	Average kinetic energy is **directly proportional to absolute temperature**

Average KE per mole = (3/2) RT
Average KE per molecule = (3/2) kT
$k = R/N_A = 1.38 \times 10^{-23}$ J K^{-1} (Boltzmann constant)
Kinetic gas equation: $PV = (1/3)\, m\, n\, U^2$

Three kinds of molecular speed

Speed	Meaning	Formula
Most probable (α)	Speed of the **largest fraction** of molecules at that temperature	$\alpha = \sqrt{(2RT/M)}$
Average (v)	Average of all the speeds	$v = \sqrt{(8RT/\pi M)}$
Root mean square (U)	Square root of the **mean of the squares** of the speeds	$U = \sqrt{(3RT/M)}$

EXAM HACK

Order of the three speeds: **rms > average > most probable**. Remember the numbers under the root: **2, 8/π, 3**. Smallest to largest: 2 < 2.55 < 3.

3.4 The gas laws

Boyle's law (1661)

At constant temperature, the volume of a fixed mass of gas is **inversely proportional to its pressure**.

$V \propto 1/P \rightarrow PV = k \rightarrow P_1V_1 = P_2V_2$
(T and n constant)

- Strictly true only for **ideal gases at very low pressure**, where molecular collisions and interactions are few.

WORKED EXAMPLE · Boyle's law

6.0 L of gas at 2.0 atm is compressed to 3.0 atm at constant temperature.

$V_2 = P_1V_1 \div P_2 = (2.0 \times 6.0) \div 3.0 =$ **4.0 L**

Pressure up, volume down.

Charles' law

At constant pressure, the volume of a fixed mass of gas is **directly proportional to its absolute temperature**.

$V \propto T \rightarrow V/T = k \rightarrow V_1/T_1 = V_2/T_2$
(P and n constant, T in kelvin)

WORKED EXAMPLE · Charles' law

4.0 L of gas at 300 K is heated to 450 K at constant pressure.

$V_2 = V_1 \times T_2 \div T_1 = 4.0 \times 450 \div 300 =$ **6.0 L**

Gay-Lussac's law

At constant volume: $P \propto T \rightarrow P_1/T_1 = P_2/T_2$

Avogadro's law

At constant T and P, the volume of a gas is **proportional to the number of moles**: $V = Kn$, so $V_1/n_1 = V_2/n_2$. Equivalent statement: **equal volumes of all gases at the same temperature and pressure contain the same number of molecules**.

Combined gas law

Boyle's law ($V \propto 1/P$) and Charles' law ($V \propto T$) combine to give $V \propto T/P$.

$P_1V_1 / T_1 = P_2V_2 / T_2$

WORKED EXAMPLE · combined gas law (book example)

10.0 L of gas at 0.70 atm and 23 °C. What is V if pressure goes to 1.4 atm and temperature to 380 K?

$T_1 = 23 + 273 = 296$ K

$V_2 = (P_1V_1T_2) \div (T_1P_2) = (0.70 \times 10.0 \times 380) \div (296 \times 1.4)$

$V_2 = 2660 \div 414.4 =$ **6.42 L**

EXAM HACK

Combined gas law steps: (1) list initial and final values, (2) **change °C to K**, (3) keep P in the same unit on both sides, (4) cross-multiply for the unknown.

3.5 Diffusion and effusion

- **Diffusion:** migration of molecules of different substances through random motion. It always goes from **higher to lower concentration**.
- **Effusion:** escape of gas molecules from a container through a tiny hole (orifice).

Graham's law: $r_1/r_2 = \sqrt{(d_2/d_1)} = \sqrt{(M_2/M_1)}$
(constant T and P; r = rate, d = density, M = molar mass)

WORKED EXAMPLE · Graham's law

How much faster does H_2 (M = 2) diffuse than O_2 (M = 32)?

$r(H_2)/r(O_2) = \sqrt{(32 \div 2)} = \sqrt{16} = \mathbf{4}$. Hydrogen is 4 times faster.

The **lighter** gas diffuses faster.

TRAP · examiners love this

Do not flip the ratio. Rate of light gas ÷ rate of heavy gas = √(heavy M ÷ light M). Quick check: your answer must say the **lighter** gas is faster.

3.6 Dalton's law of partial pressures

At constant temperature, the total pressure of a mixture of **non-reacting gases** is the **sum of the partial pressures**.

$$\mathbf{P_{total} = p_1 + p_2 + p_3 + \ldots}$$
$$\mathbf{\text{Partial pressure} = \text{mole fraction} \times \text{total pressure}}$$
$$\mathbf{PV = (n_1 + n_2 + n_3)\,RT}$$

WORKED EXAMPLE · Dalton's law

A vessel holds 2 mol N_2 and 3 mol O_2 at total pressure 5.0 atm.

Mole fraction of N_2 = 2 ÷ 5 = 0.4

$p(N_2) = 0.4 \times 5.0 = \mathbf{2.0\ atm}$; $p(O_2) = 0.6 \times 5.0 = \mathbf{3.0\ atm}$. They add to 5.0.

3.7 The ideal gas law

Combine Boyle ($V \propto 1/P$), Charles ($V \propto T$) and Avogadro ($V \propto n$):

$$\mathbf{PV = nRT}$$
$$\mathbf{R = PV/nT = (1\ atm \times 22.4\ L) \div (1\ mol \times 273\ K) = 0.082\ L\ atm\ mol^{-1}\ K^{-1}}$$

- For one mole: **PV = RT**.
- **Ideal gases** obey the ideal gas law exactly.

- Number of moles n = mass ÷ molar mass = m/M, so **PV = (m/M) RT**.
- Density d = m/V, giving **M = dRT/P** (molar mass from density).

WORKED EXAMPLE · molar mass from density (book exercise)

Compound X: density 8.9 g/L at 60 °C and 2.73 atm.

T = 60 + 273 = 333 K

M = dRT ÷ P = (8.9 × 0.082 × 333) ÷ 2.73

M = **89 g/mol**

EXAM HACK

Use **R = 0.082** with atm and litres. If pressure is in kPa or volume in cm^3, **convert first**.

3.8 Real gases and deviations

- At **high pressure and low temperature**, real gases deviate strongly from ideal behaviour.
- In an ideal gas the molecular volume is negligible, there are no attractions and collisions are perfectly elastic. In real gases the volume is appreciable, attractions exist and collisions are not perfectly elastic.
- **Compressibility factor Z = PV/nRT.** Ideal gas: **Z = 1**.
- **Negative deviation: Z < 1** (more compressible). **Positive deviation: Z > 1** (less compressible).

Factor	Effect on deviation
Nature of gas	Easily liquefiable, highly soluble gases deviate more
Pressure	More deviation at **high** pressure. CO_2 and N_2 show negative deviation at low pressure and positive deviation at high pressure
Temperature	More deviation at **low** temperature. H_2 and He always show positive deviation at 0 °C

3.9 Van der Waals equation

Van der Waals corrected the ideal gas law for the two flaws in kinetic theory.

Correction	Reason
Volume: **$V - nb$**	Molecules have real volume (extended volume), so the free space is less. b is a constant for each gas
Pressure: **$P + an^2/V^2$**	A molecule about to hit the wall is pulled inward by molecules on one side only, so it strikes with reduced speed and the measured pressure is **lower** than the ideal pressure

$(P + an^2/V^2)(V - nb) = nRT$
One mole: $(P + a/V^2)(V - b) = RT$

ALZHEIMER'S MODE

The ideal gas law pretends molecules are **dots that ignore each other**. Real molecules are **small balls that tug each other**. "b" fixes the ball size, "a" fixes the tugging.

EXAM HACK

b = **b**ulk (volume of the molecules). **a** = **a**ttraction. a goes with pressure, b goes with volume.

ALL THE HACKS FOR THIS CHAPTER

Boyle: P and V opposite (constant T). **Charles: V and T together** (constant P, kelvin).

Always add 273 to °C before any gas-law question.

Combined law: $P_1V_1/T_1 = P_2V_2/T_2$. Cross-multiply for the unknown.

Graham: the lighter gas wins. H_2 vs O_2: $\sqrt{(32 \div 2)} = 4$ times faster.

Dalton: partial pressure = mole fraction × total pressure.

M = dRT/P gives molar mass from density. R = 0.082 with atm and litres.

Real gases misbehave at high pressure and low temperature. Van der Waals: **a** = attraction, **b** = bulk.

QUICK RECALL · read this the night before

Four states: solid, liquid, gas, plasma. Matter = mass + space.

1 atm = 760 mmHg = 101.325 kPa. T(K) = °C + 273. R = 0.082 L atm mol^{-1} K^{-1}.

Kinetic theory: negligible volume, no attraction, elastic collisions, KE ∝ T.

Speeds: rms > average > most probable.

Boyle PV = k. Charles V/T = k. Gay-Lussac P/T = k. Avogadro V/n = k. Combined $P_1V_1/T_1 = P_2V_2/T_2$.

Graham: lighter gas diffuses faster, r ∝ 1/√M. Dalton: P_total = sum of partial pressures.

PV = nRT. M = dRT/P.

Real gases deviate at high P and low T. Z = PV/nRT. Van der Waals: $(P + an^2/V^2)(V - nb) = nRT$.

04 The Mole Concept and Avogadro's Number

CHAPTER 4 · READ WITH THE BOOK: PDF pages 48-52

ALZHEIMER'S MODE

You cannot count grains of rice one by one, so you buy a **cup**. Chemists cannot count atoms one by one, so they use a **mole**. A mole is just a **very big dozen**: 6.022×10^{23} pieces.

The periodic table does the hard work for you. The number under each element is the **mass of one mole** of its atoms in grams. So mass, moles and number of particles are three ways of saying the same amount. Your whole job is to **convert between them**.

4.1 The mole and Avogadro's number

- Scientists count elementary entities (electrons, atoms, ions, molecules) in **moles (mol)**.
- One mole of any particle contains exactly **6.022×10^{23}** particles. This is **Avogadro's number (N)**.
- The mole is the **reacting unit** of chemical species. It is the basis on which amounts of reactants and products are measured.

1 mole = 6.022×10^{23} particles = Avogadro's number (N)

4.2 The mole and the periodic table

The periodic table gives the **average molar mass** of each element. That number is the mass of **one mole of its atoms**. Example: copper is listed as 63.55, so

63.55 g Cu = 1 mol Cu = 6.022×10^{23} atoms of Cu

The three-way conversion

From	To	Do this
Mass (g)	Moles	Divide by molar mass
Moles	Mass (g)	Multiply by molar mass
Moles	Particles	Multiply by 6.022×10^{23}
Particles	Moles	Divide by 6.022×10^{23}
Mass (g)	Particles	Go through moles: mass → moles → particles

EXAM HACK

Moles are the **middle station**. Mass and number of particles never convert directly. Always go to moles first, then out the other side.

WORKED EXAMPLE · copper conversions (book Example 1, Cu = 63.55 g/mol)

a. Mass of 0.0892 mol Cu: $0.0892 \times 63.55 =$ **5.669 g**

b. Moles in 72.5 g Cu: $72.5 \div 63.55 =$ **1.141 mol**

c. Mass of a sample with 2.00×10^{23} Cu atoms: $(2.00 \times 10^{23} \div 6.022 \times 10^{23}) \times 63.55 =$ **21.1 g**

d. Mass of one Cu atom: $63.55 \div 6.022 \times 10^{23} =$ **1.055×10^{-22} g**

4.3 Molar mass and molecular mass

- One **formula unit** of a compound stands for **one mole** of that compound.
- 1 mol H_2SO_4 = $1 \times 6.022 \times 10^{23}$ molecules of H_2SO_4. 2 mol $MgSO_4$ = $2 \times 6.022 \times 10^{23}$ units.
- **Molar mass** = mass in **grams per mole** (g/mol).
- **Molecular mass** = mass of the same compound in **atomic mass units** (a.m.u., symbol u).

Term	Unit	Example for $(NH_4)_2SO_4$
Molecular mass	a.m.u. or u	132 u
Molar mass	g/mol	132 g/mol

WORKED EXAMPLE · molecular and molar mass of $(NH_4)_2SO_4$ (book Example 2; N = 14, H = 1, S = 32, O = 16)

$(NH_4)_2SO_4$ contains 2 N, 8 H, 1 S and 4 O.

$= (2 \times 14) + (8 \times 1) + 32 + (4 \times 16) = 28 + 8 + 32 + 64 =$ **132**

Molecular mass = **132 u**. Molar mass = **132 g/mol**.

TRAP · examiners love this

The **subscript outside the bracket multiplies everything inside**. In $(NH_4)_2SO_4$ there are 2 N and **8** H, not 4 H.

4.4 Ions and atoms inside a compound

WORKED EXAMPLE · SO_4^{2-} ions in 150 g H_2SO_4 (book Example 3a; H_2SO_4 = 98 g/mol)

1 mol H_2SO_4 contains 1 mol SO_4^{2-}, so 98 g $H_2SO_4 \leftrightarrow 6.022 \times 10^{23}$ SO_4^{2-} ions.

Number of ions = $(150 \div 98) \times 6.022 \times 10^{23} =$ **9.22×10^{23} ions**

WORKED EXAMPLE · volume of HCl that contains 2.00 mol of Cl atoms (book Example 3b; HCl density 1.18 g/mL)

1 mol HCl contains 1 mol H atoms and 1 mol Cl atoms, so 2.00 mol Cl atoms come from **2.00 mol HCl**.

Mass of 2.00 mol HCl = $2.00 \times 36.5 =$ **73.0 g**

Volume = mass ÷ density = $73.0 \div 1.18 =$ **61.9 mL**

TRAP · examiners love this

The textbook working for this example uses 71.0 g (the mass of the chlorine alone, 2×35.5) and gets 60.2 mL. The mass you convert must be the mass of **HCl**, because density is the density of HCl. If a lecturer marks with the book's 60.17 mL, show both steps clearly.

4.5 Solved exercises from the chapter

Question	Working	Answer
Mass of Ag (107.9) in 3 atoms	$3 \times 107.9 \div 6.022\times10^{23}$	5.38×10^{-22} g
Grams of Zn (65.39) in 0.54 mol	0.54×65.39	35.3 g
Mass of americium (243) in 12 atoms	$12 \times 243 \div 6.022\times10^{23}$	4.84×10^{-21} g
Moles and atoms in 20 g Ce (140.1)	$20 \div 140.1$; $\times 6.022\times10^{23}$	0.143 mol; 8.60×10^{22} atoms
Si atoms (28.09) in a 6.78 mg chip	$6.78\times10^{-3} \div 28.09 = 2.41\times10^{-4}$ mol; $\times 6.022\times10^{23}$	1.45×10^{20} atoms
Grams of C in 300 g sucrose ($C_{12}H_{22}O_{11}$ = 342)	$(144 \div 342) \times 300$	126 g
Cl atoms in 150 mL CCl_4 (d 1.589 g/mL, M = 154)	mass 238.4 g; ÷ 154 = 1.548 mol; × 4 Cl × 6.022×10^{23}	3.73×10^{24} atoms

TRAP · examiners love this

Milli means ÷ 1000. A 6.78 mg sample must be converted to **grams** (0.00678 g) before dividing by molar mass.

EXAM HACK

Mass percent of an element in a compound = (mass of the element in one mole ÷ molar mass of compound) × 100. Count atoms from the formula first.

ALL THE HACKS FOR THIS CHAPTER

Mass ↔ moles ↔ particles. Moles are always the middle station.

Mass ÷ molar mass = moles. Moles × 6.022 × 10^{23} = particles.

Mass of one atom = molar mass ÷ 6.022 × 10^{23}.

Brackets multiply: $(NH_4)_2SO_4$ has 8 H, not 4.

Milligrams ÷ 1000 first before dividing by molar mass.

QUICK RECALL · read this the night before

1 mole = 6.022 × 10^{23} particles (Avogadro's number).

Molar mass in g/mol equals the mass number on the periodic table (Cu 63.55 g/mol).

Mass ↔ moles: divide or multiply by molar mass. Moles ↔ particles: divide or multiply by 6.022 × 10^{23}.

Molecular mass is in u. Molar mass is in g/mol. Same number.

$(NH_4)_2SO_4$ = 132. Count every atom inside brackets times the subscript outside.

Mass of one atom = molar mass ÷ 6.022 × 10^{23}.

05 Relative and Average Atomic Masses

CHAPTER 5 · READ WITH THE BOOK: PDF pages 53-55

ALZHEIMER'S MODE

Atoms are far too light to weigh directly, so chemists compare them to **one standard atom**: carbon-12. Every other atom is measured as "how many times heavier than a twelfth of carbon-12".

Most elements come as a mix of isotopes, like a class with tall and short students. The mass on the periodic table is the **class average, weighted by how many students of each height there are**.

5.1 The atomic mass unit

- If we know the combining ratios of atoms in compounds, we can compare atomic masses.
- By international convention, the standard is **carbon-12**, which is given a mass of exactly **12 atomic mass units (12 u)**.
- So **one atomic mass unit (a.m.u. or u) = exactly 1/12 of the mass of a carbon-12 atom**.

5.2 Average atomic mass

The atomic mass of an element is the **weighted average of the masses of its naturally occurring isotopes**. To find it you need two things for each isotope: its **atomic mass** and its **fractional natural abundance**.

Average atomic mass = Σ (isotopic mass × fractional abundance)
Fractional abundance = % abundance ÷ 100

EXAM HACK

Steps: (1) convert each % to a decimal by ÷ 100, (2) multiply each decimal by its mass, (3) add. The answer must sit **between the lightest and the heaviest isotope**, and nearer the more abundant one.

WORKED EXAMPLE · average atomic mass of carbon (book Example 1)

Carbon-12: 12.00000 u, 98.892 %. Carbon-13: 13.00335 u, 1.108 %.

Step 1: 98.892 ÷ 100 = 0.98892 and 1.108 ÷ 100 = 0.01108

Step 2: (12.00000 × 0.98892) + (13.00335 × 0.01108)

= 11.86704 + 0.14408 = **12.011 u**

Check: close to 12 because carbon-12 is by far the most abundant.

5.3 Finding abundances from the average mass

When the average mass is given and the abundances are missing, call one fractional abundance **x** and the other **(1 − x)**. Then put it in the weighted-average equation and solve.

WORKED EXAMPLE · two isotopes of copper (book Example 2): Cu-63 = 62.9298 u, Cu-65 = 64.9278 u, average = 63.546 u

Step 1: let the fractional abundance of Cu-63 = x. Then Cu-65 = 1 − x.

Step 2: 62.9298x + 64.9278(1 − x) = 63.546

Step 3: 62.9298x + 64.9278 − 64.9278x = 63.546 → −1.998x = −1.3818

x = 1.3818 ÷ 1.998 = **0.6916**, so 1 − x = **0.3084**

Cu-63 = **69.16 %**, Cu-65 = **30.84 %**

TRAP · examiners love this

Do not forget to turn the decimal back into **percent** at the end. Check that the two percentages add to 100.

If the average is closer to one isotope's mass, that isotope must have the larger abundance. In the copper example, 63.546 is closer to 62.93 than to 64.93, so Cu-63 is the commoner one (69 %).

5.4 Finding an unknown isotope mass

The book's exercises also ask for the mass of a second isotope when the average and the first isotope's mass and abundances are given. Use the same equation, but now the **mass is the unknown**. Write (known mass × known fraction) + (unknown × its fraction) = average, then solve for the unknown.

ALL THE HACKS FOR THIS CHAPTER

1 a.m.u. = 1/12 of carbon-12.

Average mass = (mass × fraction) added for each isotope. Fractions = % ÷ 100.

The answer must lie **between** the lightest and heaviest isotope, and lean toward the commoner one.

Two isotopes, unknown abundance: call them x and (1 − x), then solve.

QUICK RECALL · read the night before

1 a.m.u. = 1/12 of the mass of a carbon-12 atom. Carbon-12 = exactly 12 u.

Average atomic mass = weighted average of the isotopes.

% to fraction: ÷ 100. Fractions of all isotopes add to 1.

Carbon worked answer: 12.011 u.

Missing abundance: let x and (1 − x), solve. Copper: Cu-63 69.16 %, Cu-65 30.84 %.

06 Stoichiometry and Chemical Equations

CHAPTER 6 · READ WITH THE BOOK: PDF pages 56-78

ALZHEIMER'S MODE

A chemical equation is a **recipe**. "2 eggs + 1 cup flour makes 1 cake" becomes "2 H_2 + 1 O_2 makes 2 H_2O". The numbers in front are how many **moles** of each ingredient the recipe uses.

Stoichiometry is just **scaling the recipe**. If you only have 5 eggs, the recipe tells you how much flour and how many cakes. The ingredient you run out of first is the **limiting reagent**, and it decides how much you can make. Learn the mole bridge once and every question in this chapter follows the same three steps.

6.1 What stoichiometry covers

Stoichiometry deals with the **quantities of chemical species in a reaction**. You need (a) the mole concept and chemical equations, (b) the **law of conservation of mass**, and (c) the periodic table. It covers empirical and molecular formulae, percentage composition, limiting reagent, percentage yield, theoretical yield, and solution concentrations (molarity, molality, % w/w, % w/v, % v/v, normality and mole fraction).

6.2 Balancing chemical equations

A **chemical equation** is a shorthand for a chemical reaction. The numbers in front of formulae are **stoichiometric coefficients**. Balance by changing coefficients only, never the subscripts.

Symbol	Meaning
(g)	gas or steam
(l)	liquid
(s)	solid
(aq)	aqueous (water) solution

Symbol	Meaning
Δ	applied heat
↓ or ppt	precipitate

WORKED EXAMPLE · balancing equations

Fe + O_2 → Fe_2O_3 (book Example 1): 4Fe + 3O_2 → **2Fe_2O_3**. Check: 4 Fe and 6 O on each side.

Propane: C_3H_8 + O_2 → CO_2 + H_2O: C_3H_8 + 5O_2 → **3CO_2 + 4H_2O**. Check: 3 C, 8 H, 10 O each side.

H_2 + O_2 → H_2O: 2H_2 + O_2 → **2H_2O**.

$SiCl_4$ + H_2O → SiO_2 + HCl: $SiCl_4$ + 2H_2O → SiO_2 + **4HCl**.

EXAM HACK

Balance in this order: **metals first, then non-metals, hydrogen second last, oxygen last**. Polyatomic ions that stay together (like SO_4^{2-}) are balanced as one block.

6.3 Percentage composition

% of element = (mass of that element in one mole ÷ molar mass of compound) × 100

WORKED EXAMPLE · % nitrogen in $Mg(NO_3)_2$ (book example)

Molar mass = 24.31 + (2 × 14.01) + (6 × 16.00) = 24.31 + 28.02 + 96.00 = **148.33 g/mol**

N in the formula: 2 atoms = 28.02 g

% N = (28.02 ÷ 148.33) × 100 = **18.89 %**

6.4 Empirical formula

The **empirical formula** is the **simplest whole-number ratio** of atoms in a compound. Example: the empirical formula of hydrogen peroxide (H_2O_2) is **HO**.

Compound	Molecular formula	Empirical formula
Hydrogen peroxide	H_2O_2	HO
Hydrazine	N_2H_4	NH_2
Glucose	$C_6H_{12}O_6$	CH_2O
Benzene	C_6H_6	CH

From percentage composition

1. Treat percent as grams (take 100 g)
2. Divide each mass by its atomic mass to get moles
3. Divide every mole value by the SMALLEST
4. If not whole numbers, multiply to make them whole

WORKED EXAMPLE · empirical formula of phenol (book): 76.57 % C, 6.49 % H, 17.00 % O

Moles: C = 76.57 ÷ 12 = 6.38; H = 6.49 ÷ 1 = 6.49; O = 17.00 ÷ 16 = 1.0625

Divide by the smallest (1.0625): C = 6.00, H = 6.11, O = 1.00

Ratio = 6 : 6 : 1, so the empirical formula is **C_6H_6O**.

From combustion (chemical) analysis

Burn the compound; collect CO_2 and H_2O. All the **carbon** goes to CO_2 (12 g C in every 44 g CO_2). All the **hydrogen** goes to H_2O (2 g H in every 18 g H_2O). **Oxygen is found by difference.**

WORKED EXAMPLE · an alcohol (book): 0.10 g burnt gives 0.1910 g CO_2 and 0.1172 g H_2O

Mass of C = (12 ÷ 44) × 0.1910 = **0.0521 g**

Mass of H = (2 ÷ 18) × 0.1172 = **0.0130 g**

Mass of O = 0.10 − (0.0521 + 0.0130) = **0.0349 g**

Moles: C = 0.0521 ÷ 12 = 0.00434; H = 0.0130 ÷ 1 = 0.0130; O = 0.0349 ÷ 16 = 0.00218

Divide by 0.00218: C = 2, H = 6, O = 1. Empirical formula **C_2H_6O**.

TRAP · examiners love this

Oxygen in the combustion method is **never** read from the CO_2 and H_2O, because some oxygen came from the air. Always get it by difference.

Do not round ratios too early. 1.5 means multiply everything by 2. 1.33 means multiply by 3.

6.5 Molecular formula

The molecular formula shows the full number of each atom in a molecule. It is a **whole-number multiple** of the empirical formula.

Molecular formula = (empirical formula) × n
n = molecular mass ÷ empirical formula mass

WORKED EXAMPLE · fruit sugar (book): molecular mass 180.2, empirical formula CH_2O

Empirical formula mass = 12 + 2 + 16 = 30

n = 180.2 ÷ 30 = 6.007 = **6**

Molecular formula = $(CH_2O)_6$ = **$C_6H_{12}O_6$**

WORKED EXAMPLE · NO_2 with molecular mass 92.0 (book exercise)

Empirical mass = 14 + 32 = 46

n = 92 ÷ 46 = 2, so the molecular formula is **N_2O_4**.

6.6 Calculations from balanced equations

Mole bridge: given → moles → (mole ratio) → moles of wanted → required unit
1. Balance the equation
2. Convert the given quantity to moles
3. Use the coefficients as a ratio
4. Convert to the unit asked for

WORKED EXAMPLE · moles of CO_2 from 4.30 mol of benzene (book)

$2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O$

2 mol C_6H_6 gives 12 mol CO_2

4.30 mol C_6H_6 × (12 ÷ 2) = **25.8 mol CO_2**

WORKED EXAMPLE · mass of oxygen from $KClO_3$ (type of book exercise)

$2KClO_3 \rightarrow 2KCl + 3O_2$. Take 2.50 g $KClO_3$ (122.5 g/mol).

Moles $KClO_3$ = 2.50 ÷ 122.5 = 0.0204 mol

Moles O_2 = 0.0204 × 3/2 = 0.0306 mol

Mass O_2 = 0.0306 × 32 = **0.98 g**

6.7 Limiting reagent and percent yield

- The **limiting reagent** is the reactant used up first. It **limits the amount of product**. The other reactant is in **excess**.
- When all of the limiting reagent has reacted, the reaction stops even though excess reagent remains.
- **Theoretical yield:** the amount of product calculated from the balanced equation, based on the limiting reagent.
- **Actual yield** is always less than the theoretical, because of side reactions, impurities, losses and weighing errors.

% yield = (actual yield ÷ theoretical yield) × 100

WORKED EXAMPLE · limiting reagent (book): 5.0 mol Ba and 12.0 mol H_2O. $Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$

1 mol Ba reacts with 2 mol H_2O.

5.0 mol Ba needs 5.0 × 2 = **10 mol H_2O**. There are 12 mol, so water is more than enough.

Limiting reagent = Ba. Excess reagent = H_2O (2 mol left over).

WORKED EXAMPLE · ammonia (book): 25.0 g N_2 and 5.0 g H_2. $N_2 + 3H_2 \rightarrow 2NH_3$

N_2 = 28 g/mol and $3H_2$ = 6 g. So 28 g N_2 needs 6 g H_2.

25.0 g N_2 needs (6 ÷ 28) × 25.0 = **5.357 g H_2**. Only 5.0 g is present, so **H_2 is limiting**.

Product from the limiting reagent: 6 g H_2 gives 34 g NH_3, so 5.0 g H_2 gives (34 ÷ 6) × 5.0 = **28.33 g NH_3**.

If only 24.0 g NH_3 is actually collected: % yield = (24.0 ÷ 28.33) × 100 = **84.7 %**.

EXAM HACK

Limiting reagent test: work out how much of reactant B is needed to use up **all** of reactant A. If you have less than that, **B is limiting**. If you have more, A is limiting. Always do the product calculation from the **limiting** one.

TRAP · examiners love this

Never calculate the product from the reactant that is in excess. It always gives too big an answer.

6.8 Stoichiometry in solution

Solution = solute + solvent. Concentration can be given in many ways:

Term	Symbol	Formula
Molarity	M	moles of solute ÷ litres of **solution**
Molality	m	moles of solute ÷ kg of **solvent**
Percent by mass	% (m/m)	(mass of solute ÷ mass of solution) × 100
Percent (w/v), (v/v)		grams or mL of solute per 100 mL solution
Mole fraction	x_A	n_A ÷ (n_A + n_B)
Normality	N	Equivalents of solute per litre

TRAP · examiners love this

Molarity uses litres of solution. Molality uses kilograms of solvent. M for Molarity, m for molality: capital is the bigger quantity (whole solution), small m is the smaller one (solvent only).

WORKED EXAMPLE · molarity (book): 13.5 g NaOH in 2500 mL of solution (NaOH = 40)

Moles = 13.5 ÷ 40 = 0.3375 mol. Volume = 2500 mL = **2.5 L**

M = 0.3375 ÷ 2.5 = **0.135 M**

WORKED EXAMPLE · ions in solution (book)

0.50 M K_2SO_4 → K_2SO_4 → $2K^+ + SO_4^{2-}$, so $[K^+]$ = 2 × 0.50 = **1.0 M** and $[SO_4^{2-}]$ = **0.50 M**.

2.0 M HCl → $H^+ + Cl^-$, so $[H^+]$ = **2.0 M** and $[Cl^-]$ = **2.0 M**.

WORKED EXAMPLE · moles and mass from volume (book)

Moles of Na^+ in 15 cm^3 of 0.8 M NaCl: 0.8 × 15 ÷ 1000 = 0.012 mol NaCl = **0.012 mol Na^+**

Mass of HNO_3 in 300 mL of 0.30 M (63 g/mol): 0.30 × 0.300 = 0.09 mol; 0.09 × 63 = **5.67 g**

WORKED EXAMPLE · percent concentration (book)

1.5 g NaCl in 45 g water: total mass = 46.5 g. % (m/m) = (1.5 ÷ 46.5) × 100 = **3.23 %**

2.0 g ethanol in 110.0 g water: (2.0 ÷ 112.0) × 100 = **1.786 %**

WORKED EXAMPLE · molality (book)

25.0 g I_2 (254 g/mol) in 200 g CCl_4: moles = 25.0 ÷ 254 = 0.098 mol. Solvent = 0.200 kg. m = 0.098 ÷ 0.200 = **0.49 mol/kg**

1.65 M H_2SO_4 (98 g/mol), density 1.29 g/mL: 1000 mL weighs 1290 g and holds 1.65 × 98 = 161.7 g H_2SO_4. Water = 1290 − 161.7 = 1128.3 g = 1.1283 kg. m = 1.65 ÷ 1.1283 = **1.46 mol/kg**

WORKED EXAMPLE · mole fraction (book): 30 % (m/m) aqueous glucose (180 g/mol)

In 100 g solution: 30 g glucose = 30 ÷ 180 = 0.167 mol; 70 g water = 70 ÷ 18 = 3.89 mol

x(glucose) = 0.167 ÷ (0.167 + 3.89) = **0.041**

Standard solution, stock solution and dilution

- A **standard solution** has an exactly known concentration.
- A **stock solution** is a highly concentrated standard solution used to make dilute ones.

$M_1V_1 = M_2V_2$ (stock = 1, dilute = 2)

WORKED EXAMPLE · dilution (book): prepare 500 mL of 2.0 M acetic acid from 16.0 M stock

$V_1 = M_2V_2 \div M_1 = (2.0 \times 500) \div 16.0 =$ **62.5 mL**

Pipette 62.5 mL of stock into a 500 mL volumetric flask, make up to the mark with distilled water, then shake to mix.

EXAM HACK

Dilution always **lowers** concentration and **raises** volume. So $M_1 > M_2$ and $V_1 < V_2$. If your V_1 is bigger than V_2, you have swapped them.

ALL THE HACKS FOR THIS CHAPTER

Balancing order: metals → non-metals → H → O.

Empirical formula: % → grams → ÷ atomic mass → ÷ smallest → whole numbers.

Combustion analysis: C from CO_2 (× 12/44), H from H_2O (× 2/18), **O by difference**.

Molecular formula = empirical × n, n = molecular mass ÷ empirical mass.

Limiting reagent: work out how much of B reacts with all of A. If B is short, B limits.

Molarity: litres of Solution. molality: kg of sOlvent.

Dilution: $M_1V_1 = M_2V_2$. V_1 must be smaller than V_2.

QUICK RECALL · read this the night before

Balance with coefficients only. Symbols: (g), (l), (s), (aq), Δ, ↓.

% composition = (element mass ÷ molar mass) × 100.

Empirical: divide by atomic mass, then by the smallest. Combustion: C from CO_2 (12/44), H from H_2O (2/18), O by difference.

Molecular = empirical × n; n = molecular mass ÷ empirical mass.

Mole bridge: balance → moles → ratio → unit.

Limiting reagent controls the yield. % yield = actual ÷ theoretical × 100.

M = mol/L of solution. m = mol/kg of solvent. x = n_A/(n_A + n_B).

Dilution: $M_1V_1 = M_2V_2$.

07 Acids, Bases and Salts

CHAPTER 7 · READ WITH THE BOOK: PDF pages 79-93

ALZHEIMER'S MODE

There are **three ways to define an acid**, each one wider than the last. Arrhenius: acid gives H^+ in water. Brønsted-Lowry: acid **gives away** a proton. Lewis: acid **takes** an electron pair. Same chemistry, bigger and bigger lens.

pH is just a **compression tool**. Instead of writing 0.00001 M, you write pH 5. The smaller the pH, the more acid. Buffers are the **shock absorbers** of chemistry: they soak up a little added acid or base without the pH jumping.

Salts are what is left when an acid and a base meet. Whether the salt is neutral, acidic or basic depends on **which side was stronger**.

7.1 Arrhenius definition

- **Acid:** any species which, dissolved in water, dissociates to give **hydrogen ions (H^+) as the only positive ions**. Examples: HCl → $H^+ + Cl^-$, $H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$, HNO_3, CH_3COOH.
- A free proton does not exist alone in water. It attaches to water: **$HCl + H_2O \rightarrow H_3O^+ + Cl^-$**.
- **Base:** any species which in water gives **hydroxide ions (OH^-) as the only negative ions**. Example: $NaOH \rightarrow Na^+ + OH^-$. Ammonia: $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$.

Type of acid	Meaning	Examples
Monoprotic	Gives **one** proton per molecule	HCl, HNO_3
Polyprotic	Gives **more than one** proton per molecule	H_2CO_3 (2), H_3PO_4 (3)

7.2 Brønsted-Lowry definition

An **acid is a proton (H^+) donor** and a **base is a proton acceptor**.

- In $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, HCl is the acid (donates a proton to water). Water is the base (accepts it).
- In the reverse reaction, H_3O^+ is a potential acid and Cl^- a potential base.
- H_3O^+ is the **conjugate acid** of H_2O. Cl^- is the **conjugate base** of HCl.

Acid	Base	Conjugate acid	Conjugate base
HCl	H_2O	H_3O^+	Cl^-
HCl	NH_3	NH_4^+	Cl^-
HNO_2	F^-	HF	NO_2^-

EXAM HACK

Conjugate pair = **differ by exactly one H^+**. To find the conjugate base: **remove H^+**. To find the conjugate acid: **add H^+**. Examples: conjugate base of HCN is **CN^-**, of OH^- is **O^{2-}**; conjugate acid of CO_3^{2-} is **HCO_3^-**, of Br^- is **HBr**.

Autoionisation (self-ionisation)

When one species acts as **both** a Brønsted acid and a base. In general, two neutral molecules of the same substance form a **cation-anion pair**.

$H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$
$NH_3 + NH_3 \rightleftharpoons NH_4^+ + NH_2^-$
$CH_3COOH + CH_3COOH \rightleftharpoons CH_3COOH_2^+ + CH_3COO^-$

7.3 Lewis definition

- **Lewis acid:** a species that **accepts a pair of electrons**.
- **Lewis base:** a species that **donates a pair of electrons** to form a covalent bond.
- Examples: $H^+ + OH^- \rightarrow H_2O$. $BCl_3 + NH_3 \rightarrow Cl_3B{\leftarrow}NH_3$ (BCl_3 is the acid, NH_3 the base).

- The Lewis concept is **more universal** than Arrhenius and Brønsted-Lowry because **electrons are involved in all chemical reactions**.

Concept	Acid	Base
Arrhenius	Gives H^+ in water	Gives OH^- in water
Brønsted-Lowry	Proton **donor**	Proton **acceptor**
Lewis	Electron-pair **acceptor**	Electron-pair **donor**

> **EXAM HACK**
>
> **Arrhenius = Aqueous**. **Brønsted = proton Bus** (donates the seat). **Lewis = Lone-pair Loan**: the base lends the lone pair.

7.4 Strength of acids and bases

- **As an acid gets stronger, its conjugate base gets weaker.**
- Strong acids in the book's order (strongest first): $HClO_4$, H_2SO_4, HI, HBr, HCl, HNO_3, H_3O^+, then weaker ones such as HSO_4^- and H_3PO_4.

Levelling and differentiating solvents

- With water as the reference base, HCl, HNO_3 and $HClO_4$ all appear **100 % ionised**. Their differences are removed. This is the **levelling effect**.
- If the solvent is pure ethanoic (glacial acetic) acid, a **much weaker base than water**, the extent of ionisation differs noticeably. It is a **differentiating solvent**.

7.5 Salts

A **salt** is the compound formed when ionisable protons of an acid have been replaced by metallic ions. There are **five types**: normal, acid, basic, double and complex.

Type	How it forms	Example
Normal	**All** replaceable H^+ replaced. Usually neutral to litmus (though Na_2CO_3 and $AlCl_3$ hydrolyse to give alkaline or acidic solutions)	Na_2SO_4, NaCl
Acid	**Partial** replacement of H^+, when metal ions are too few. Still has an H^+ that can be replaced, so acts as both salt and acid. A dibasic acid forms **one** acid salt, a tribasic acid forms **two**	$NaHSO_4$, KH_2PO_4, K_2HPO_4
Basic	A base reacts with **too little acid**. Seen as a normal salt with some oxygen or hydroxyl in place of anion. Turns red litmus blue, reacts with excess acid to give normal salt + water	SbOCl (from $SbCl_3$ + H_2O)
Double	Two simple salts combined in stoichiometric ratio. Gives **three kinds of ions** (two cations, one anion) and shows the reactions of both parent salts	$(NH_4)_2SO_4 \cdot FeSO_4 \cdot 6H_2O$
Complex	Contains a **complex ion**: a metal ion surrounded by ligands (NH_3, NO_2^-, SO_3^{2-}, CN^-). Stays together in solution	$K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-}$

- **Alums** are double salts with the formula $M^+M^{3+}(SO_4)_2 \cdot 12H_2O$, where M^{3+} is Al^{3+} or Fe^{3+}. Example: **potassium alum $KAl(SO_4)_2 \cdot 12H_2O$**, used to treat cotton fibres so they take dyes better.
- $[Fe(CN)_6]^{4-}$ is named **hexacyanoferrate(II)** ion.

TRAP · examiners love this

Do not mix up **acid salt** (still has a replaceable H, e.g. $NaHSO_4$) and **acidic salt solution** (a normal salt whose solution is acidic after hydrolysis, e.g. NH_4Cl). One is a type of salt, the other is a property of its solution.

7.6 The pH concept

$pH = -\log [H^+]$
$pOH = -\log [OH^-]$
$pH + pOH = 14$ ($K_w = [H^+][OH^-] = 1.0 \times 10^{-14}$ at 25 °C)
$[H^+] = 10^{(-pH)}$

WORKED EXAMPLE · pH calculations (book Example 1)

a. $[H^+] = 0.030$ M: pH = −log(0.030) = **1.5**

b. $[OH^-] = 2.0 \times 10^{-2}$ M: pOH = −log(0.020) = 1.7. pH = 14 − 1.7 = **12.3**

c. pH = 8.5: $[H^+] = 10^{-8.5}$ = **3.16×10^{-9} M**

EXAM HACK

pH below 7 acidic, 7 neutral, above 7 basic. **Each pH unit is a factor of 10.** pH 3 is 10 times more acidic than pH 4 and 100 times more than pH 5.

TRAP · examiners love this

If a question gives **$[OH^-]$**, you get pOH first, then subtract from 14. Many students stop at pOH and give the wrong answer.

7.7 Buffer solutions

- A **buffer** is a solution of a **weak acid and its salt** (or a weak base and its salt). **Both** components must be present.
- It **resists changes in pH** when a small amount of acid or base is added.
- In other words a buffer is an **acid-base conjugate pair**: a weak acid with its conjugate base, or a weak base with its conjugate acid.

If you add...	What happens in an acetic acid / acetate buffer
Base (OH^-)	The acid part neutralises it: $CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$
Acid (H^+)	The conjugate base soaks it up: $CH_3COO^- + H^+ \rightarrow CH_3COOH$

The acetate ions come from the **complete dissociation of sodium acetate**: $CH_3COONa \rightarrow CH_3COO^- + Na^+$. The buffer is written CH_3COONa/CH_3COOH or CH_3COO^-/CH_3COOH.

Henderson equation (acid buffer):
pH = pKa + log ([salt] ÷ [acid])
pKa = −log Ka

WORKED EXAMPLE · buffer pH (book): 1.0 M CH_3COOH and 1.0 M CH_3COONa; Ka = 1.8×10^{-5}

Before HCl: $pH = -\log(1.8\times10^{-5}) + \log(1.0 \div 1.0) = 4.74 + 0 =$ **4.74**

After adding 0.10 mol HCl to 1 L: $H^+ + CH_3COO^- \rightarrow CH_3COOH$. So $[CH_3COOH] = 1.0 + 0.1 = 1.1$ M and $[CH_3COO^-] = 1.0 - 0.1 = 0.9$ M.

$pH = 4.74 + \log(0.9 \div 1.1) =$ **4.66**

The pH barely moved. That is the whole point of a buffer.

Buffer range

Buffer range: pH = pKa ± 1

Buffer pair	pKa	Range
CH_3COO^- / CH_3COOH	4.74	3.74 to 5.74
HCO_3^- / H_2CO_3	6.38	5.38 to 7.38
CO_3^{2-} / HCO_3^-	10.32	9.32 to 11.32

WORKED EXAMPLE · moles of sodium acetate for a pH 6 buffer (book exercise): 1.0 L of 0.2 M acetic acid, Ka = 1.8×10^{-5}

pKa = 4.74. $6 = 4.74 + \log(\text{salt} \div 0.2) \rightarrow \log(\text{salt} \div 0.2) = 1.26$

$\text{salt} \div 0.2 = 10^{1.26} = 18$, so salt = $18 \times 0.2 \approx$ **3.6 mol** sodium acetate.

EXAM HACK

Henderson equation memory: **salt over acid, add to pKa**. If salt = acid, the log is zero and **pH = pKa**.

7.8 Salt hydrolysis

- **Hydrolysis** is any reaction involving a water molecule as a reactant.
- Weak acids and bases dissociate only slightly. If Ka or Kb is 1.0×10^{-4} or less, the amount dissociated (x) is negligible against the starting concentration. This is the **fair approximation**.
- The **common ion effect:** dissociation of a weak acid (or base) is **discouraged** if a salt of that acid (or base) is added.

Salt of	Hydrolysis	Example	Solution is
Strong base + weak acid	**Anionic**: the anion takes a proton from water	$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$	Alkaline
Weak base + strong acid	**Cationic**: the cation gives a proton to water	$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$	Acidic
Strong acid + strong base	No hydrolysis	NaCl	Neutral

Anionic: Kh = Kw / Ka
Cationic: Kh = Kw / Kb
$Kw = Ka \times Kb = 1.0 \times 10^{-14}$

EXAM HACK

Strong wins. The stronger parent decides the solution. Strong base + weak acid → basic. Strong acid + weak base → acidic.

7.9 Solubility product (Ksp)

- The **solubility product** is the product of the molar concentrations of the ions, each raised to the power of its coefficient.
- **The smaller the Ksp, the less soluble the compound.**
- $AgCl(s) \rightleftharpoons Ag^+ + Cl^-$; $Ksp = [Ag^+][Cl^-] = 1.6 \times 10^{-10}$ (book value).

Ion product Q

Comparison	Meaning
Q < Ksp	**Unsaturated** solution. No precipitate
Q = Ksp	**Saturated** solution (equilibrium)
Q > Ksp	**Supersaturated**. A precipitate forms until the ion product falls back to Ksp

- **Molar solubility:** moles of solute in one litre of a saturated solution (mol/L) at 25 °C.
- **Solubility:** grams of solute in one litre of saturated solution (g/L) at 25 °C.

WORKED EXAMPLE · Ksp from molar solubility (book): molar solubility of Ag_2SO_4 = 1.5 × 10^{-2} mol/L

$Ag_2SO_4(s) \rightleftharpoons 2Ag^+ + SO_4^{2-}$. One mole dissolves to give 2 mol Ag^+ and 1 mol SO_4^{2-}.

$[Ag^+] = 2 \times 1.5\times10^{-2} = 3.0\times10^{-2}$ M; $[SO_4^{2-}] = 1.5\times10^{-2}$ M

$Ksp = [Ag^+]^2[SO_4^{2-}] = (3.0\times10^{-2})^2 \times 1.5\times10^{-2} =$ **1.35×10^{-5}**

WORKED EXAMPLE · Ksp from solubility in g/L (book exercise): $CaSO_4$, 0.67 g/L, M = 136.29

Molar solubility = 0.67 ÷ 136.29 = 4.9 × 10^{-3} mol/L

$CaSO_4 \rightleftharpoons Ca^{2+} + SO_4^{2-}$, so $[Ca^{2+}] = [SO_4^{2-}] = 4.9\times10^{-3}$ M

$Ksp = (4.9\times10^{-3})^2 =$ **2.4×10^{-5}**

WORKED EXAMPLE · molar solubility from Ksp (book exercise): CuI, Ksp = 5.1 × 10^{-12}

$CuI \rightleftharpoons Cu^+ + I^-$. If solubility is s, $Ksp = s \times s = s^2$.

$s = \sqrt{(5.1\times10^{-12})} =$ **2.3×10^{-6} mol/L**

Predicting precipitation

WORKED EXAMPLE · will $BaSO_4$ precipitate? (book): 200 cm^3 of 0.0040 M $BaCl_2$ + 600 cm^3 of 0.0080 M K_2SO_4. Ksp($BaSO_4$) = 1.1 × 10^{-10}

The only possible precipitate is $BaSO_4$. (KCl does not precipitate.)

Moles Ba^{2+} = 0.200 L × 0.0040 = 8.0×10^{-4} mol. Total volume = 800 cm^3 = 0.800 L, so [Ba^{2+}] = **1.0×10^{-3} M**

Moles SO_4^{2-} = 0.600 × 0.0080 = 4.8×10^{-3} mol; [SO_4^{2-}] = 4.8×10^{-3} ÷ 0.800 = **6.0×10^{-3} M**

Q = $(1.0\times10^{-3})(6.0\times10^{-3}) = 6.0\times10^{-6}$. This is far **greater than Ksp**, so **a precipitate forms**.

TRAP · examiners love this

After mixing, **volumes add**, so each ion's concentration is **diluted**. Always find the concentrations in the final total volume before comparing Q with Ksp.

ALL THE HACKS FOR THIS CHAPTER

Three definitions: Arrhenius = H^+/OH^- in water. Brønsted = proton donor/acceptor. Lewis = electron-pair acceptor/donor.

Conjugate pair differs by one H^+. Strong acid → weak conjugate base.

pH + pOH = 14. Given [OH^-]? Find pOH first.

Each pH unit = ×10.

Buffer: weak acid + its salt. **pH = pKa + log(salt/acid)**. Salt = acid → pH = pKa.

Hydrolysis: the stronger parent wins. Strong base + weak acid → basic.

Q vs Ksp: Q > Ksp → precipitate forms. After mixing, dilute the concentrations first.

QUICK RECALL · read this the night before

Arrhenius: H^+ / OH^- in water. Brønsted-Lowry: proton donor / acceptor. Lewis: electron-pair acceptor / donor.

Conjugate pair differs by one H^+. Stronger acid, weaker conjugate base.

Levelling solvent: water (HCl, HNO_3, $HClO_4$ look equal). Differentiating: glacial acetic acid.

Five salts: normal, acid, basic, double (alum), complex.

pH = $-\log[H^+]$; pH + pOH = 14.

Buffer = weak acid + its salt. Henderson: pH = pKa + log(salt/acid). Range = pKa ± 1.

Hydrolysis: strong base + weak acid → alkaline; strong acid + weak base → acidic. Kh = Kw/Ka.

Ksp: smaller = less soluble. Q < Ksp unsaturated, Q = Ksp saturated, Q > Ksp precipitate.

08 Chemical Bonding and Intermolecular Forces

CHAPTER 8 · READ WITH THE BOOK: PDF pages 94-99

ALZHEIMER'S MODE

Atoms are like people who want a **full set**. For atoms, the full set is **8 outer electrons**, the noble gas arrangement. There are only three ways to get there: **give** electrons away (ionic), **share** them (covalent), or let **one partner pay for both** (coordinate).

Metals have spare electrons, non-metals need electrons, so metal + non-metal usually means ionic. Two non-metals share, so covalent. After that, molecules still tug on each other weakly. Those tugs are the **intermolecular forces**.

8.1 Chemical bond and the octet rule

- A **chemical bond** is the **electrostatic force of attraction between negatively charged electrons and positively charged nuclei**. It holds bonded atoms or ions together.
- Atoms react to reach a stable configuration, the **octet** (the noble gas electron arrangement). The book credits this idea to **Lewis, G. N.**
- **Octet rule:** when an atom forms a bond by **gaining, losing or sharing valence electrons**, its electron configuration becomes the same as that of the **noble gas before or after it** in the periodic table.
- The rule works for many atoms but **not always**. The major criterion for a reaction to occur is the **energetics** of the reaction.
- Three methods of reaching an octet: **electrovalent**, **covalent** and **coordinate** bonding.

8.2 Electrovalent (ionic) bonding

- The **electrostatic attraction between two oppositely charged ions**.

- Involves the **complete transfer of one or more electrons** from one atom to another.
- Forms between atoms that readily **lose** electrons (Groups I and II) and atoms that readily **gain** them (Groups VII and VI).
- Example: in **NaCl**, sodium transfers one electron to chlorine. Na becomes Na^+ with the **neon** configuration, Cl becomes Cl^- with the **argon** configuration.

Properties of electrovalent compounds

Property	Reason
Made of ions in **crystal lattices** (a lattice can be treated as a macromolecule)	Positive and negative ions stack in a regular pattern
High melting and boiling points	Strong forces between the ions
Soluble in **polar solvents**	Polar solvents have a **high dielectric constant**, which helps separate and solvate individual ions
React **fast**	The ions just collide to form new compounds. No bonds to break

8.3 Covalent bonding

- Exists when the **electronegativity difference** between two elements is **small**. Formed by **sharing electrons**.
- Brought about by the **overlap of singly filled orbitals**.
- **Oxygen molecule (O_2):** each oxygen atom shares **two** electrons with the other (a double bond).
- **Carbon dioxide (CO_2):** carbon has 4 of 8 outer electrons and each oxygen has 6 of 8. By sharing, every atom gets a full outer shell.
- Other examples: H_2, Cl_2, CCl_4.
- Two types: the **normal** covalent bond and the **coordinate covalent bond**.

Coordinate (dative) covalent bond

A pair of electrons **donated by one atom** is shared between two atoms. Example: the ammonium ion **NH_4^+**, where the lone pair of NH_3 is donated to H^+.

Properties of covalent compounds

Property	Reason
Made of **discrete molecules**	Only weak van der Waals forces between molecules
Gases, liquids or **soft solids** with **low** melting and boiling points	Weak forces between molecules
Soluble in **non-polar** solvents	Those have a low dielectric constant
React **slowly**	Old bonds must be broken to form new ones, which needs high energy

	Ionic	Covalent
Bond formed by	Transfer of electrons	Sharing of electrons
Units	Crystal lattice of ions	Discrete molecules
Melting point	High	Low
Dissolves in	Polar solvent (water)	Non-polar solvent
Reaction speed	Fast	Slow

EXAM HACK

Like dissolves like. Ionic and polar substances dissolve in polar solvents like water. Covalent and non-polar substances dissolve in non-polar solvents like CCl_4.

8.4 Dipole and polarity

- Two atoms of the **same element** joined by a covalent bond: **no dipole**.
- Two **different** atoms: usually a **dipole**, because one has greater affinity for electrons.
- **Water:** oxygen has greater affinity for electrons than hydrogen, so the molecule has a **dipole moment** and is a **polar solvent**.

- **CCl_4:** the four Cl atoms pull equally, so the dipoles cancel. **No dipole moment**, so it is a **non-polar solvent**.
- **Dipole moment** = the product of the **distance separating equal charges of opposite sign** and the **magnitude of the charge**.

> **TRAP · examiners love this**
>
> CCl_4 **has** polar C-Cl bonds but the **molecule** is non-polar because the four pulls cancel. The question is about the molecule, not the bond.

8.5 Metallic bonding

- Holds atoms together in **solid metals**.
- Metal **cations** sit in a **crystal lattice** in a definite geometric pattern. The **valence electrons are free** to move throughout the crystal.
- These free electrons explain the **high thermal and electrical conductivity** of metals.

> **ALZHEIMER'S MODE**
>
> Metal = a neat grid of positive ions sitting in a **sea of free electrons**. Electrons can flow, so heat and electricity pass through easily.

8.6 Intermolecular forces

Between all particles, especially covalent molecules, there are very weak attractions called **intermolecular forces**. They are **much weaker than full covalent or ionic bonds**.

Hydrogen bonding

- Occurs when **hydrogen** is bonded to a very electronegative atom: **fluorine, oxygen or nitrogen**.
- The electronegative atom gets a **negative partial charge** and hydrogen a **positive partial charge**.
- The H is strongly attracted to the F, O or N of a neighbouring molecule.
- Examples: **water** and **methanol**.

EXAM HACK

F-O-N with H: remember **"FON"** (phone). Hydrogen bonds need H attached to F, O or N.

Van der Waals forces

- Relatively **weak** forces attracting **neutral molecules** to one another, in gases, liquids and solidified gases.
- Solids held by them have **lower melting points** and are **softer** than those held by ionic, covalent or metallic bonds.
- Three causes: (i) **permanent electric dipoles** that align, (ii) **polarisation induced by nearby** polar or non-polar molecules, (iii) the attraction between molecules such as the **noble gas argon** and the organic liquid **benzene**.

Bond / force	Strength
Ionic, covalent, metallic	Strong
Hydrogen bond	Weaker, but stronger than van der Waals
Van der Waals	Weakest

ALL THE HACKS FOR THIS CHAPTER

Metal + non-metal = ionic (transfer). Non-metal + non-metal = covalent (share).

Ionic: lattice, high mp, fast reactions, dissolves in polar solvents.
Covalent: molecules, low mp, slow, dissolves in non-polar solvents.

Like dissolves like.

CCl_4 has polar bonds but is a non-polar molecule (pulls cancel).

Hydrogen bond = H with F, O or N. "**FON**".

Strength order: ionic/covalent/metallic > hydrogen bond > van der Waals.

QUICK RECALL · read this the night before

Bond = electrostatic attraction between electrons and nuclei. Octet = noble gas configuration.

Three ways: electrovalent (transfer), covalent (share), coordinate (one atom donates the pair, e.g. NH_4^+).

Ionic: lattice, high mp, soluble in polar solvents, fast. Covalent: molecules, low mp, soluble in non-polar, slow.

Dipole moment = distance × charge. Water polar, CCl_4 non-polar.

Metals: cations in a sea of free electrons, conduct heat and electricity.

Hydrogen bond: H with F, O or N. Van der Waals: weak, between neutral molecules (argon, benzene).

09 Chemical Equilibrium and Thermodynamics

CHAPTER 9 · READ WITH THE BOOK: PDF pages 100-113

ALZHEIMER'S MODE

Equilibrium is **two people running up a down escalator at the same speed**. They look like they stay in one place, but both are working hard. The reaction goes forward and backward at the **same rate**, so the amounts stop changing, but nothing has stopped.

Le Chatelier is the **grumpy neighbour rule**: whatever you do to the system, it does the opposite to cancel you out.

Thermodynamics asks **two questions**. Does the reaction give out or take in heat (enthalpy)? Does the mess increase (entropy)? ΔG combines both answers and tells you if the reaction will happen on its own.

9.1 Chemical equilibrium

- **Chemical equilibrium** is the state where the **concentrations of reactants and products remain constant with time**. At molecular level it is **not static**, it is **dynamic**.
- A **reversible reaction** can be made to go in either direction by changing the conditions. Examples: steam on red-hot iron ($3Fe + 4H_2O \rightleftharpoons Fe_3O_4 + 4H_2$) and heating ammonium chloride ($NH_4Cl \rightleftharpoons NH_3 + HCl$).
- As a reversible reaction proceeds, concentrations level up and become constant. The **forward rate equals the backward rate**. That is equilibrium.

Properties of a system at equilibrium
1. It is **dynamic**
2. Rates are **equal in opposite directions**
3. The **concentration** of each reactant and product is **constant**
4. **ΔG = 0**

Properties of a system at equilibrium

5. The system **resists disturbance**

9.2 The equilibrium constant

For A + B ⇌ C + D, the forward rate is $R_1 = k_1[A][B]$ and the backward rate is $R_2 = k_2[C][D]$. At equilibrium $R_1 = R_2$, so

$K = k_1/k_2 = [C][D] / [A][B]$
For $aA + bB \rightleftharpoons cC + dD$: $K = [C]^c[D]^d / [A]^a[B]^b$

- **Products on top, reactants on the bottom**, each raised to the power of its coefficient.
- K may or may not have units. Work out the units from the expression.

WORKED EXAMPLE · writing K and its units (book Question 1)

$2NO_2 \rightleftharpoons N_2O_4$: $K = [N_2O_4] / [NO_2]^2$. Units: mol/L ÷ (mol/L)² = **L/mol**.

$N_2 + 3H_2 \rightleftharpoons 2NH_3$: $K = [NH_3]^2 / ([N_2][H_2]^3)$. Units: (mol/L)² ÷ (mol/L)⁴ = **L²/mol²**.

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$: $K = [PCl_3][Cl_2] / [PCl_5]$. Units: **mol/L**.

Ester + water: $CH_3CH_2CO_2H + CH_3CH_2OH \rightleftharpoons CH_3CH_2CO_2CH_2CH_3 + H_2O$: $K = [ester][H_2O] / ([acid][alcohol])$. Powers cancel top and bottom, so **no units**.

EXAM HACK

Units of K: **(total power on top) − (total power on bottom)**. If equal, K has **no units**.

9.3 Homogeneous and heterogeneous equilibrium

Type	Meaning	Example
Homogeneous	All substances in the **same phase**	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
Heterogeneous	Components in **different phases**	$PbI_2(s) \rightleftharpoons Pb^{2+}(aq) + 2I^-(aq)$

TRAP · examiners love this

In K for a heterogeneous equilibrium, **pure solids and pure liquids are left out** (their concentration is taken as 1). For $Ca(OH)_2(s) \rightleftharpoons Ca^{2+} + 2OH^-$, **$K = [Ca^{2+}][OH^-]^2$**. Never write the solid.

9.4 Le Chatelier's principle (law of mobile equilibrium)

If a system at equilibrium is disturbed, it **shifts its position to counter the disturbance**. The main factors are **temperature, pressure and concentration**.

Disturbance	Shift	Equilibrium constant K
Add a reactant	Away from that species (to the **product** side)	Unchanged
Remove a species	Toward that species (side where it was removed)	Unchanged
Increase pressure (reduce volume)	Side with **fewer gas molecules**	Unchanged
Decrease pressure	Side with **more gas molecules**	Unchanged
Raise temperature	Direction that **absorbs heat** (endothermic direction)	**Changes**
Add a **catalyst**	**No shift.** Equilibrium reached faster	Unchanged

Concentration

For $PCl_5 \rightleftharpoons PCl_3 + Cl_2$: adding PCl_5 shifts equilibrium **right**, using up Cl_2 and making more PCl_3. Removing a species makes the reaction fill the gap, so it favours the side where the species was removed.

Pressure

- Pressure changes come from volume changes.
- **$N_2 + 3H_2 \rightleftharpoons 2NH_3$:** 4 moles on the left, 2 moles on the right. Increasing pressure drives the reaction to the **right** (fewer molecules). Lowering pressure favours decomposition of NH_3 back to N_2 and H_2.
- **$N_2 + O_2 \rightleftharpoons 2NO$:** 2 moles on each side, so pressure has **no effect**.

Temperature

- Exothermic (ΔH negative): heat is a **product**. Endothermic (ΔH positive): heat is a **reactant**.
- **$H_2 + I_2 \rightleftharpoons 2HI$ + heat** is exothermic. Raising the temperature shifts the equilibrium **left** (less HI). Lowering the temperature gives **more HI**.

Catalyst

A catalyst has **no effect on the position of equilibrium**. It only helps equilibrium to be **attained in a shorter time**.

EXAM HACK

Temperature: treat **heat as a reactant or product**. Exothermic → heat on the right → more heat pushes it **left**. Pressure: count **gas moles each side**, go to the **smaller** side when pressure rises.

TRAP · examiners love this

A catalyst does **not** change K and does **not** shift equilibrium. It speeds up both directions equally. Only a **temperature** change changes the value of K.

9.5 Thermodynamics: basic terms

- **Thermochemistry:** heat effects that accompany chemical reactions. A subfield of **thermodynamics**.
- **Thermo** = heat. **Dynamics** = motion or change. Chemical thermodynamics studies the **energy transformations** in chemical or physical changes.
- **System:** part of the universe chosen for study. **Surroundings:** everything outside it that it interacts with.

System	Exchanges	Example
Open	**Energy and matter**	Beaker of hot coffee (loses heat and water vapour)
Closed	**Energy only**, no matter	Stoppered flask of hot coffee
Isolated	**Neither**	Hot coffee in an insulated container (approximately)

9.6 Energy, work and the first law

- **Energy** is the capacity to do work. **Work** is done when a **force acts through a distance**: work = force × distance. The unit is the **joule**.
- Work and heat are not forms of energy. They are **processes** by which the energy of a system is changed.
- **Kinetic energy** = energy of motion (½mv²). **Potential energy** = energy from condition, position or composition (attraction or repulsion between objects).
- **Thermal energy** is the kinetic energy of random molecular motion and is proportional to temperature.

First law: ΔE = q + w
Energy is neither created nor destroyed. It only changes form.

Sign convention	Meaning
q > 0	Heat **absorbed** by the system
q < 0	Heat **given off** by the system

Sign convention	Meaning
w > 0	Work done **on** the system
w < 0	Work done **by** the system
ΔE > 0	More energy enters than leaves

- For an **isolated system**, no heat or work is exchanged, so **ΔE = 0** and energy is constant.

WORKED EXAMPLE · first law (type of book Question 2)

A system absorbs 600 kJ of heat and does 450 kJ of work on its surroundings.

q = +600 kJ, w = −450 kJ (work done **by** the system is negative)

ΔE = q + w = 600 − 450 = **+150 kJ**

If instead 450 kJ of work is done **on** the system, ΔE = 600 + 450 = **+1050 kJ**.

TRAP · examiners love this

The textbook line "w is the work done by the system" and its sign rule ("w > 0 if work is done on the system") do not match. Use the sign rule: **work on the system is positive, work by the system is negative**.

9.7 Enthalpy

H = E + PV
ΔH = ΔE + PΔV (constant pressure)
ΔH negative → exothermic. ΔH positive → endothermic.

	Exothermic	Endothermic
Heat	Given off to surroundings	Absorbed from surroundings
ΔH	**Negative**	**Positive**
Isolated system	Temperature rises	Temperature falls

9.8 Second law, entropy and Gibbs free energy

- **Second law:** all physical and chemical changes occur so that **some energy disperses**. Concentrated energy becomes diffuse and random.
- **Entropy (S)** measures the **degree of disorderliness** of a system. Raising the temperature **increases** entropy.
- The entropy of the universe **increases** in every change: $\Delta S(\text{universe}) > 0$.
- The entropy of an **isolated system never spontaneously decreases**: $\Delta S(\text{isolated}) \geq 0$.

Gibbs free energy: $G = H - TS$
$\Delta G = \Delta H - T\Delta S$ (T in kelvin)

ΔG	Meaning
$\Delta G < 0$	**Spontaneous** in the forward direction
$\Delta G > 0$	**Not spontaneous** forward. Needs outside energy. Spontaneous in the reverse direction
$\Delta G = 0$	**At equilibrium**. No driving force either way

WORKED EXAMPLE · free energy change (book Question 3): ΔH = 4500 J, ΔS = 12 J/K at 27 °C

$T = 27 + 273 =$ **300 K**

$T\Delta S = 300 \times 12 = 3600$ J

$\Delta G = \Delta H - T\Delta S = 4500 - 3600 =$ **+900 J**

ΔG is positive, so the reaction is **not spontaneous** at 27 °C.

TRAP · examiners love this

Convert **°C to K** before multiplying by ΔS. And keep ΔH and TΔS in the same unit (both J or both kJ).

EXAM HACK

ΔH negative and ΔS positive → ΔG always negative (always spontaneous). ΔH positive and ΔS negative → never spontaneous. In the other two cases, **temperature decides**.

9.9 Activation energy and the Arrhenius equation

- **Activation energy (Ea)** is the **energy barrier** that must be overcome before a reaction can occur. It is the difference between the energy of the reactant molecules and that of the **activation complex** (transition state) at the top of the barrier.
- The number of molecules that have this energy decides the **reaction rate**.

Arrhenius: k = A e^(−Ea/RT)
Linear form: ln k = ln A − Ea/(RT)
Plot ln k against 1/T: straight line, slope = −Ea/R, intercept = ln A

TRAP · examiners love this

The slope is **−Ea/R** (negative), because ln k falls as 1/T rises. The line of best fit slopes **downward**.

- In an **energy profile diagram** the activation complex is the **rate-determining step**.
- Exothermic: products lie **lower** in energy than reactants. Endothermic: products lie **higher**, energy is absorbed.

9.10 Types of enthalpy change

Name	Meaning
Formation (ΔHf)	Heat evolved or absorbed when **one mole of a compound is formed from its elements**
Combustion	Heat evolved when **one mole** of a substance is **completely oxidised**
Solution	Heat change when one mole of solute dissolves completely in a large excess of water
Neutralisation	Heat evolved when **one equivalent** of acid is neutralised by one equivalent of base in **dilute solution**
Dilution	Heat change when a solution containing one mole of solute is diluted from one concentration to another
Precipitation	Heat evolved in the precipitation of one mole of a **sparingly soluble** substance from dilute solutions
Hydration	Heat change when one mole of an anhydrous (or partly hydrated) salt combines with the required moles of water to form a specific hydrate
Fusion	Enthalpy change converting 1 mole **solid → liquid** at its melting point (mostly endothermic)
Vaporisation	1 mole **liquid → gas** at its boiling point
Sublimation	1 mole **solid → vapour directly**

EXAM HACK

Fusion = **f**reezing/melting (solid ↔ liquid). Vaporisation = boiling. Sublimation = **skips** the liquid.

ALL THE HACKS FOR THIS CHAPTER

Equilibrium: equal rates, constant amounts, not stopped.

K = products ÷ reactants (powers = coefficients). **Leave out solids and pure liquids.**

Only temperature changes K. A catalyst changes speed, not position.

Pressure up → side with fewer gas moles. Heat up → endothermic direction (treat heat as a reactant).

ΔG = ΔH − TΔS. Negative = spontaneous, zero = equilibrium. T in kelvin.

ΔH negative = exothermic. ΔH positive = endothermic.

Arrhenius plot: ln k vs 1/T, slope = −Ea/R.

QUICK RECALL · read this the night before

Equilibrium: dynamic, equal rates, constant concentrations, ΔG = 0, resists disturbance.

K = [products]^{c} ÷ [reactants]^{a}. Leave out pure solids and liquids.

Le Chatelier: opposes any change. Catalyst: no shift, faster. Only temperature changes K.

Open exchanges energy + matter; closed energy only; isolated neither.

ΔE = q + w. H = E + PV. ΔH negative exothermic, positive endothermic.

Entropy = disorder, increases in the universe. ΔG = ΔH − TΔS. ΔG < 0 spontaneous, = 0 equilibrium, > 0 not spontaneous.

k = Ae^(−Ea/RT); ln k vs 1/T slope = −Ea/R.

Ten enthalpy types: formation, combustion, solution, neutralisation, dilution, precipitation, hydration, fusion, vaporisation, sublimation.

10 Chemical Kinetics and Nuclear Chemistry

CHAPTER 10 · READ WITH THE BOOK: PDF pages 114-127

ALZHEIMER'S MODE

Kinetics is the speedometer of chemistry. Thermodynamics tells you if a reaction can happen. Kinetics tells you **how fast**, and how it happens step by step (the mechanism).

Nuclear chemistry is chemistry inside the nucleus. Normal reactions only shuffle electrons. Nuclear reactions change **what the atom is**. An unstable nucleus throws out a particle or energy to become more stable. The only rule when writing nuclear equations is: **top numbers add up, bottom numbers add up**.

10.1 Chemical kinetics

- **Chemical kinetics** deals with **reaction rate**: the rate at which a chemical process occurs. It also sheds light on the **reaction mechanism** (how the reaction occurs).
- Rate depends on factors such as **concentration, temperature and catalyst**.
- Two main goals: (1) analyse the **sequence of elementary steps** (the mechanism), (2) find the **absolute rate** of the reaction and its elementary steps.

10.2 Rate law

The **rate law** is the equation relating the **reaction rate to the concentration of the reactants**. Rates are written **negative for reactants** (they decrease) and **positive for products** (they increase). For A → P:

rate = −Δ[A]/Δt = Δ[P]/Δt
For aA + bB → cC + dD: rate = k [A]^m [B]^n
k = rate constant. m, n = orders. Overall order = m + n

TRAP · examiners love this

The exponents m and n are found **by experiment**. They are **not necessarily equal to the coefficients a and b** in the balanced equation.

Exponent	Meaning	If concentration doubles
0 (zero order)	Rate does **not depend** on that reactant	Rate unchanged
1 (first order)	Rate **directly proportional** to concentration	Rate **doubles**
2 (second order)	Rate proportional to the **square**	Rate **quadruples**

10.3 Classification of reactions

By molecularity

Molecularity is the number of molecules that **simultaneously take part** in an elementary step.

Type	Meaning	Example	Rate
Unimolecular	One molecule (decomposition or rearrangement)	$I_2 \rightarrow 2I$	rate = k[A]; usually first order
Bimolecular	Collision of two molecules (same or different)	$H_2 + I_2 \rightarrow 2HI$	rate = k[A][B]; **second order**
Termolecular	Collision of three molecules	$2NO + H_2 \rightarrow N_2O + H_2O$	rate = k[A][B][C]; **third order**

- Higher molecularity is possible in theory, but **simultaneous collision of more than three molecules is improbable**. Reactions of higher molecularity than three are **not known in practice**. Reactions with many molecules proceed in **stages**.

TRAP · examiners love this

Molecularity is for an **elementary step** and is a whole number (1, 2, 3). **Order** comes from the rate law and can be 0, 1, 2, 3 or a fraction. Do not swap the two words.

By order of reaction

Order of reaction is the **sum of the exponents** of the concentrations in the rate law. The classes are zero, first, second and third order.

10.4 Integrated rate equations

Order	Rate law	Integrated equation	Half-life
Zero	rate = k	$[A]_0 - [A]t = kt$	$[A]_0 \div 2k$
First	rate = k[A]	$\ln[A]t - \ln[A]_0 = -kt$ or $[A]t = [A]_0$ e^(−kt)	$\ln 2 \div k = 0.693 \div k$
Second	rate = $k[A]^2$	$1/[A]t = kt + 1/[A]_0$	$1 \div (k[A]_0)$
Third	rate = $k[A]^3$	$1/[A]t^2 - 1/[A]_0^2 = 2kt$	$3 \div (2k[A]_0^2)$

- **Half-life (t½)** is the time for **half of the original concentration** to be used up.
- For first order: put $[A]t = [A]_0/2$ into the first-order equation. $\ln 2 = kt_{½}$, so **$t_{½} = \ln 2 / k = 0.693 / k$**.
- For **first order**, $t_{½}$ does **not depend on the starting concentration**. For second order it does.

Plot	Gives a straight line for	Slope
ln[A] against t	First order	−k
1/[A] against t	Second order	+k
[A] against t	Zero order	−k

EXAM HACK

Which graph is straight? **Zero:** [A] vs t. **First:** ln[A] vs t. **Second:** 1/[A] vs t. The numbers 0, 1, 2 line up with [A], ln[A], 1/[A].

General equation (order n): $(1/(n-1)) \times (1/[A]t^{(n-1)} - 1/[A]_0^{(n-1)}) = kt$

10.5 Nuclear chemistry: discovery of radioactivity

- **1896: Henri Becquerel** found that uranium crystals emitted rays that fogged photographic plates. He called them **uranic rays**. The rays came from changes inside the **nuclei** of uranium atoms, which were unstable.
- **Marie Sklodowska Curie** (doctoral work in France) found two new elements, **polonium** (named after her native **Poland**) and **radium**. She renamed uranic rays **radioactivity**.
- The **Nobel Prize in physics** for discovering radioactivity was shared by **Marie Curie, Becquerel and her husband Pierre Curie**.
- **Radioactivity** = spontaneous emission of particles and/or electromagnetic radiation by **unstable nuclei** (radionuclides). The radiation is alpha particles, beta particles and gamma rays.
- All elements **heavier than bismuth** (and some lighter) are naturally radioactive.
- Nuclear reactions change the particles in the **nucleus**, so they cause **transmutation** of one element into another isotope or element. Chemical reactions only form molecules.
- The number of **protons** defines the element, so changing protons changes the element.

10.6 Types of radiation

Type	Symbol	What it is	Change in Z	Change in mass number A
Alpha	$^{4}_{2}\mathrm{He}$ or α	2 protons + 2 neutrons (helium nucleus)	**−2**	**−4**
Beta	$^{0}_{-1}\mathrm{e}$ or β	A neutron becomes a proton + an electron, and the electron leaves	**+1**	**0**
Gamma	γ	Electromagnetic energy from the nucleus	0	0
Positron	$^{0}_{+1}\mathrm{e}$	Same mass as an electron, **opposite charge**	**−1**	0
Electron capture		Nucleus captures an inner-shell electron	**−1**	0

- **Gamma:** no particles are emitted, so it does not cause transmutation by itself. It often comes **with** alpha and beta decay. X-rays emitted during beta decay of **cobalt-60** are a common example.

Nuclear equation (book)
Alpha: $^{238}_{92}\mathrm{U} \rightarrow {}^{234}_{90}\mathrm{Th} + {}^{4}_{2}\mathrm{He}$
Beta: carbon-14 → nitrogen-14 + β particle
Positron: carbon-11 → boron-11 + positron
Electron capture: $^{81}_{37}\mathrm{Rb} + {}^{0}_{-1}\mathrm{e} \rightarrow {}^{81}_{36}\mathrm{Kr}$

EXAM HACK

Alpha: **Z down 2, A down 4**. Beta: **Z up 1, A same**. Gamma: **nothing changes**. Positron: Z down 1, A same. Electron capture: Z down 1, A same.

10.7 Balancing nuclear equations

The nuclide is written $^{A}_{Z}X$: **A = mass number** (protons + neutrons), **Z = atomic number** (protons). In every nuclear equation **both mass numbers and atomic numbers are conserved**: the totals on the left equal the totals on the right.

WORKED EXAMPLE · missing nuclide (book): $^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{90}_{38}Sr + ? + 3\ {}^{1}_{0}n$

Left: mass = 235 + 1 = 236, atomic number = 92 + 0 = 92.

Right (known parts): mass = 90 + 3 = 93, atomic number = 38.

Missing: mass = 236 − 93 = **143**, atomic number = 92 − 38 = **54**.

The missing nuclide is $^{143}_{54}\mathbf{Xe}$ (xenon-143).

WORKED EXAMPLE · alpha emission of americium-243 (book Question 4a)

Alpha: A − 4, Z − 2. Am has Z = 95.

$^{243}_{95}Am \rightarrow {}^{239}_{93}Np + {}^{4}_{2}He$

10.8 Half-life and decay constant

- **Half-life** is the time required for **50 % of the radioactive nuclides to decay**. Half of whatever is left decays every half-life.
- **Decay constant (λ)** is the **instantaneous rate of decay per atom**: atoms disintegrating per second ÷ atoms present.
- **$\lambda = 0.693 / t_{½}$**.
- Half-life measures the **stability of the nucleus**. Decay is **unaffected by temperature, pressure or physical state** (solid, liquid or gas).

WORKED EXAMPLE · bismuth-210 (book): half-life 5 days, start with 100 g

After 5 days: 50 g Bi left (and 50 g Tl).

After 10 days: 25 g Bi left (and 75 g Tl).

Rule: after n half-lives, fraction left = $(½)^{n}$.

WORKED EXAMPLE · fraction remaining (book Questions 3 and 4b)

Half-life 60 days, after 120 days: 120 ÷ 60 = 2 half-lives. Fraction = $(½)^2$ = **1/4**.

Iron-59 (anaemia diagnosis), half-life 45 days, after 90 days: 2 half-lives, fraction = **1/4**.

10.9 Natural and artificial radioactivity; fission and fusion

Term	Meaning
Natural radioactivity	**Spontaneous** disintegration of unstable nuclei, e.g. $^{238}U \rightarrow {}^{234}Th + {}^{4}He$
Artificial radioactivity	**Induced** nuclear reaction when nuclei are bombarded with **high-velocity particles** (neutrons, protons, alpha particles), e.g. $^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$
Nuclear fission	**Splitting** of heavy, unstable nuclides into smaller, more stable nuclides plus neutrons. **Highly exothermic**
Nuclear fusion (thermonuclear)	**Combination** of small nuclei into larger, more stable nuclei. Needs **very high temperatures** to start, but is highly exothermic, produces **minimal radioactive waste** and uses readily available fuel (hydrogen)
Nuclear transmutation	**Induced** nuclear reactions where nuclei are bombarded with nucleons (protons, neutrons) or other nuclei to make a different element. Example: nitrogen-14 → oxygen-17 using alpha particles from radium

- **Fusion releases more energy than fission.**
- **Fusion** needs a very high temperature to start. Fission does not.

TRAP · examiners love this

Fusion needs the high temperature, fission does not. The textbook page has a typo in this line. **Fusion = Fuse = join small nuclei. Fission = Fracture = split big nuclei.**

10.10 Applications of radioactivity

Field	Use
Medicine	**Cobalt-60** destroys cancer cells (radiotherapy)
Medicine	**Iodine-131** studies thyroid gland function (production of **thyroxin**) and can treat cancer
Medicine	**Gamma rays** sterilise syringes and other medical apparatus
Medicine	**Barium-131** detects bone tumours
Agriculture	**Gamma rays** sterilise packages and foods to prevent spoilage
Agriculture	Gamma rays destroy dangerous **pests on cattle** by sterilising them
Agriculture	**Phosphorus-32** in fertiliser shows how a plant uses phosphorus

ALL THE HACKS FOR THIS CHAPTER

Order tells how rate changes: 0 none, 1 doubles, 2 quadruples when concentration doubles.

Exponents come from experiment, not from the balanced equation.

Straight-line plots: zero [A] vs t, first ln[A] vs t, second 1/[A] vs t.

First-order half-life = 0.693 / k, independent of concentration.

Alpha: A −4, Z −2. **Beta:** Z +1. **Gamma:** no change. Add up top and bottom numbers on each side.

Fraction left = (½)^(time ÷ half-life).

Fission splits, Fusion joins (needs very high temperature).

QUICK RECALL · read this the night before

Kinetics = rate + mechanism. Rate = $k[A]^m[B]^n$. Exponents are from experiment.

Order 0/1/2: rate unchanged / doubles / quadruples when concentration doubles.

Molecularity 1, 2, 3 = uni-, bi-, termolecular. Bimolecular → second order. Termolecular → third.

First order: $\ln[A]t - \ln[A]_0 = -kt$, $t½ = 0.693/k$. Second: $1/[A]t = kt + 1/[A]_0$, $t½ = 1/(k[A]_0)$.

Becquerel 1896, Marie Curie (polonium after Poland, radium), Pierre Curie, shared the Nobel Prize in physics.

Alpha −4/−2. Beta A same, Z +1. Gamma no change. Conserve A and Z.

Fraction left = (½)^(time ÷ half-life). $\lambda = 0.693/t½$.

Fission splits, fusion joins (needs very high temperature). Co-60 cancer, I-131 thyroid, P-32 fertiliser.

11 Colligative Properties

CHAPTER 11 · READ WITH THE BOOK: PDF pages 128-135

ALZHEIMER'S MODE

Colligative means "depends on the crowd, not the individuals". It does not matter whether the dissolved particles are sugar or urea, big or small. Only **how many** particles there are matters.

Add a pinch of non-volatile solute and four things change: the liquid **evaporates less** (vapour pressure goes down), **freezes later** (freezing point drops), **boils later** (boiling point rises), and it **pulls in water** through a membrane (osmotic pressure). One idea, four effects.

11.1 What are colligative properties?

- Physical properties that depend **only on the number (concentration) of solute particles**: **vapour pressure, boiling point, freezing point and osmotic pressure**.
- The **nature** of the solute (ionic, molecular mass, size) does **not** affect them.
- The difference from the pure solvent depends only on the number of **non-volatile** solute particles.
- Uses: find the **concentration of solute particles**, the **molecular weight** of the solute, and the **purity** of solute samples. Also used in car radiators and managing road-surface snow.

11.2 Vapour pressure lowering (Raoult's law)

F. M. Raoult (1887): the vapour pressure of a solution equals the mole fraction of the solvent times the vapour pressure of the pure solvent.

$$P(\text{solution}) = X(\text{solvent}) \times P^\circ(\text{solvent})$$
$$\Delta P = P^\circ - P(\text{solution}) = X(\text{solute}) \times P^\circ(\text{solvent})$$

- A **non-volatile solute lowers** the vapour pressure.
- Raoult's law holds strictly for an **ideal solution**, though it works for dilute non-ideal ones (X(solvent) > 0.97).
- In an ideal solution, the solute-solvent forces are the same as in the pure components.

WORKED EXAMPLE · vapour pressure (book Example 1): 95.0 g glycerine ($C_3H_8O_3$, 92.1 g/mol) in 190.0 g ethanol (C_2H_5OH, 46.0 g/mol) at 40 °C. P°(ethanol) = 135.3 torr

Moles glycerine = 95.0 ÷ 92.1 = 1.03 mol. Moles ethanol = 190.0 ÷ 46.0 = 4.13 mol.

X(ethanol) = 4.13 ÷ (4.13 + 1.03) = 4.13 ÷ 5.16 = **0.80**

P(solution) = 0.80 × 135.3 = **108.2 torr**

ΔP = 135.3 − 108.2 = **27.1 torr**. Check: X(glycerine) = 0.20, and 0.20 × 135.3 = 27.1 ✔

WORKED EXAMPLE · benzene and toluene (book exercise 1): equal mole fractions, P° = 95.1 and 28.4 mm Hg

Each has X = 0.5. Both are volatile, so add the partial pressures.

P = (0.5 × 95.1) + (0.5 × 28.4) = 47.55 + 14.2 = **61.75 mm Hg**

WORKED EXAMPLE · sucrose and urea in water (book exercise 3, worked in the book): 0.25 mol sucrose + 75.0 g urea in 1 kg water at 20 °C; P°(water) = 17.5 mm Hg

Urea: 75.0 ÷ 60.06 = 1.25 mol. Water: 1000 ÷ 18.02 = 55.5 mol.

Total solute = 0.25 + 1.25 = 1.50 mol.

X(water) = 55.5 ÷ (55.5 + 1.50) = **0.974**

P = 0.974 × 17.5 = **17.0 mm Hg**

TRAP · examiners love this

Use the **mole fraction of the SOLVENT** to find the new vapour pressure. Use the mole fraction of the **SOLUTE** to find the **drop** (ΔP). Mixing these up is the common mistake.

11.3 Freezing point depression

- The freezing point is the temperature at which the liquid and solid phases have the **same vapour pressure**. It is where the first solid appears.
- A solution freezes at a **lower** temperature than the pure solvent. The difference is **proportional to the molality** of the non-volatile solute.
- **K_f** is the molal freezing point depression constant (cryoscopic constant), the depression when the solute is 1.0 m.

$\Delta T_f = T_f(\text{solution}) - T_f(\text{solvent}) = -K_f \times m$
m = moles of solute ÷ kg of solvent
K_f (water) = 1.86 °C m^{-1}

WORKED EXAMPLE · freezing point (book Example 2): 25.0 g sucrose (342.3 g/mol) in 100.0 g water

Moles = 25.0 ÷ 342.3 = 0.073 mol. Solvent = 0.1000 kg.

Molality = 0.073 ÷ 0.1000 = **0.73 m**

ΔT_f = 1.86 × 0.73 = 1.36 °C

Freezing point = 0.0 − 1.36 = **−1.36 °C**

WORKED EXAMPLE · iodine in chloroform (book exercise 4): 92.1 g I_2 (253.8 g/mol) in 800.0 g $CHCl_3$, K_f = 4.68 °C/m, normal freezing point −63.5 °C

Moles = 92.1 ÷ 253.8 = 0.363 mol. Molality = 0.363 ÷ 0.8000 = 0.454 m.

ΔT_f = 4.68 × 0.454 = 2.12 °C

New freezing point = −63.5 − 2.12 = **−65.6 °C**

TRAP · examiners love this

The depression is a **drop**, so you **subtract** it from the normal freezing point. Do not add it.

11.4 Boiling point elevation

- A liquid boils when its **vapour pressure equals the external pressure**.
- A non-volatile solute **lowers** the vapour pressure, so a **higher temperature** is needed to make it boil.
- **K_b** is the molal boiling point elevation constant (ebullioscopic constant). For water **$K_b = 0.512\ °C\ m^{-1}$**.
- Application: antifreeze in car radiators.

$$\Delta T_b = T_b(\text{solution}) - T_b(\text{solvent}) = K_b \times m$$

WORKED EXAMPLE · boiling point of the sucrose solution above

m = 0.73 mol/kg

$\Delta T_b = 0.512 \times 0.73 =$ **0.37 °C**

The solution boils at 100 + 0.37 = **100.37 °C**

EXAM HACK

Freezing goes Down, boiling goes Up. Use **K_f for freezing** (1.86 for water) and **K_b for boiling** (0.512 for water). K_f is bigger, so freezing changes more than boiling.

11.5 Osmotic pressure

- **Osmosis** is the passage of **solvent molecules from low solute concentration to high solute concentration** through a **semipermeable membrane** (parchment paper, animal skins, etc.).
- **Osmotic pressure (π)** is the pressure that must be applied to the **more concentrated side** to stop the solvent moving across.
- Osmosis occurs in living things: cell membranes, the lining of the digestive tract, walls of blood vessels, and water movement in plant cells.
- Osmotic pressure is useful for finding the **molar masses of macromolecules** (proteins, polymers), which have high molar mass and low molarity but measurable osmotic pressure.

van't Hoff: $\pi = (n/V)\,RT = M\,R\,T$
π = osmotic pressure, n = moles of solute, V = volume in litres, T in kelvin, M = molarity
$R = 0.08206$ L atm mol^{-1} K^{-1}

WORKED EXAMPLE · molar mass of haemoglobin (book): 20.0 g in 1 L of solution, π = 5.92 torr at 22 °C

$T = 22 + 273 = 295$ K. $\pi = 5.92 \div 760 = 7.8 \times 10^{-3}$ atm.

$M = \pi \div RT = 7.8\times10^{-3} \div (0.08206 \times 295) =$ **3.2×10^{-4} mol/L**

Molar mass = mass ÷ moles = $20.0 \div 3.2\times10^{-4} \approx$ **6.2×10^{4} g/mol** (the book gets 62 500 g/mol using rounded values)

TRAP · examiners love this

Convert **torr to atm** (÷ 760) before using R = 0.08206 L **atm**. Convert **°C to K**.

The final step is **mass ÷ moles**, not the other way around.

Colligative property	Change	Equation
Vapour pressure	Decreases	$P = X(\text{solvent})\,P°$
Freezing point	Decreases	$\Delta T_f = -K_f m$
Boiling point	Increases	$\Delta T_b = K_b m$
Osmotic pressure	Pressure needed to stop osmosis	$\pi = MRT$

ALL THE HACKS FOR THIS CHAPTER

Colligative = number of particles, not their identity.

Freezing goes Down ($K_f = 1.86$). Boiling goes Up ($K_b = 0.512$).

Vapour pressure: new P uses mole fraction of the **solvent**; the drop uses the **solute**.

Molality = mol ÷ kg of solvent.

π = MRT. Torr ÷ 760 for atm, °C + 273 for K. Molar mass = mass ÷ moles.

QUICK RECALL · read this the night before

Colligative properties depend only on the number of non-volatile solute particles: vapour pressure, boiling point, freezing point, osmotic pressure.

Raoult (1887): $P = X(\text{solvent})\, P°$. $\Delta P = X(\text{solute})\, P°$.

Freezing point goes down: $\Delta T_f = -K_f m$ (water 1.86). Boiling point goes up: $\Delta T_b = K_b m$ (water 0.512).

Molality = mol solute ÷ kg solvent.

Osmosis: solvent moves from dilute to concentrated through a semipermeable membrane. π = MRT. Used for molar masses of big molecules.

Uses: solute concentration, molecular weight, purity; road-snow management, car radiators.

12 Reduction-Oxidation Reactions

CHAPTER 12 · READ WITH THE BOOK: PDF pages 136-150

ALZHEIMER'S MODE

Redox is a trade of electrons. One substance loses them (oxidation), the other gains them (reduction). It is always **a pair**: you cannot lose an electron unless someone else takes it.

Use **OIL RIG**: **O**xidation **I**s **L**oss, **R**eduction **I**s **G**ain (of electrons). The substance that is oxidised is the **reducing agent** (it reduces the other one). The substance that is reduced is the **oxidising agent**.

Oxidation numbers are the **bookkeeping** that lets you see the trade, even when no ions form. Balancing is just making sure **the electrons given equal the electrons taken**.

12.1 What is a redox reaction?

- A **redox reaction** is one in which **electrons are transferred** from one reactant to another. One reactant is **reduced** and the other **oxidised**, **at the same time**.
- Examples in the book: **combustion of fossil fuels**, the action of **bleaching agents**, and the reaction between **carbon and dichromate solution** in determining soil organic carbon.
- Reduction and oxidation are **complementary**: one completes the other.

12.2 Five ways to express redox

#	Reduction is…	Oxidation is…	Example
1	**Removal of oxygen**	**Addition of oxygen**	$CuO + H_2 \rightarrow Cu + H_2O$ (CuO reduced); $2Mg + O_2 \rightarrow 2MgO$ (Mg oxidised)
2	**Addition of hydrogen**	**Removal of hydrogen**	$2H_2 + O_2 \rightarrow 2H_2O$ (O_2 reduced); $H_2S + SO_2 \rightarrow S + H_2O$ (H_2S oxidised)
3	**Decrease** in electronegative atoms around an element	**Increase** in electronegative atoms	$2FeCl_3 \rightarrow 2FeCl_2 + Cl_2$ ($FeCl_3$ reduced); $PbCl_2 \rightarrow PbCl_4$ (oxidised)
4	**Gain of electrons**	**Loss of electrons**	$Cl_2 + 2e^- \rightarrow 2Cl^-$; $Na \rightarrow Na^+ + e^-$
5	**Decrease** in oxidation number	**Increase** in oxidation number	$4Na + O_2 \rightarrow 2Na_2O$: Na 0 → +1 (oxidised), O 0 → −2 (reduced)

EXAM HACK

OIL RIG covers #4. **LEO the lion says GER**: **L**oss of **E**lectrons is **O**xidation, **G**ain of **E**lectrons is **R**eduction. For oxidation number: **ox goes up, red goes down**.

TRAP · examiners love this

In $CuO + H_2 \rightarrow Cu + H_2O$ the substance that is **reduced** is CuO, but the **reducing agent** is H_2. The agent is the **other** one. Oxidant = the one **reduced**. Reductant = the one **oxidised**.

An equation like $Cl_2 + 2e^- \rightarrow 2Cl^-$ is a **half-reaction**: it shows only the reduction or only the oxidation.

12.3 Oxidising and reducing agents

	Reducing agent (reductant)	Oxidising agent (oxidant)
Does what	Causes reduction of another species	Causes oxidation of another species
Itself	**Oxidised** (loses electrons)	**Reduced** (gains electrons)
Mostly	**Electron donors**: metals, some non-metals like C and H, compounds like CO, $LiAlH_4$, $NaBH_4$	**Electron acceptors**: non-metals such as O_2, F_2, Cl_2
Strong ones		Acidified $KMnO_4$, acidified $K_2Cr_2O_7$, concentrated H_2SO_4, concentrated HNO_3

- The oxidising power of halogens **decreases down Group 7**. So a lower member cannot displace a higher one: **Cl_2 + KF → no reaction**.
- Redox properties are **relative**. Some substances act as either oxidant or reductant depending on the **other reactant**. Example: in $2H_2S + SO_2 \rightarrow 3S + 2H_2O$, **$SO_2$ is the oxidant** (H_2S is a better reductant). In $2SO_2 + O_2 \rightarrow 2SO_3$, **$SO_2$ is the reductant** (O_2 is a better oxidant).

WORKED EXAMPLE · identify the oxidant and reductant (book)

a. $2H_2S + SO_2 \rightarrow 3S + 2H_2O$: S goes from −2 (in H_2S) to 0, and from +4 (in SO_2) to 0. **SO_2 is the oxidant**, **H_2S is the reductant**.

b. $2SO_2 + O_2 \rightarrow 2SO_3$: S goes +4 → +6, O goes 0 → −2. **O_2 is the oxidant**, **SO_2 is the reductant**.

c. $2Mg + O_2 \rightarrow 2MgO$: Mg 0 → +2, O 0 → −2. **O_2 is the oxidant**, **Mg is the reductant**.

12.4 Tests for oxidising and reducing agents

Test for a REDUCING agent	Colour change
Decolourises acidified $KMnO_4$: $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	**Purple → colourless**
Changes acidified $K_2Cr_2O_7$: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	**Orange → green**
Changes aqueous Fe^{3+} salt: $Fe^{3+} + e^- \rightarrow Fe^{2+}$	**Brown → green**

Test for an OXIDISING agent	What you see
Liberates **iodine** from aqueous KI: $2I^- \rightarrow I_2 + 2e^-$	Iodine **blackens starch** solution or paper
Liberates sulphur from H_2S: $2H_2S + SO_2 \rightarrow 3S + 2H_2O$	**Whitish-yellow** sulphur deposit
Liberates **chlorine** from concentrated HCl (used to prepare chlorine gas)	Chlorine gas
Acts on a freshly made Fe^{2+} solution, acidified with dilute H_2SO_4, and warmed: $Fe^{2+} \rightarrow Fe^{3+} + e^-$	**Pale green → pale-yellow brown**

EXAM HACK

Reducing-agent colour tests: Purple, Orange, Brown all lose their colour: purple → colourless, orange → green, brown (Fe^{3+}) → green. **Oxidising-agent tests:** iodine from KI (starch turns blue-black), sulphur from H_2S, chlorine from HCl, $Fe^{2+} \rightarrow Fe^{3+}$.

12.5 Rules for oxidation numbers

#	Rule	Examples
1	**Free (uncombined) element** = **0**	H_2, Br_2, K, Xe, O_2, P_4, S_8
2	**Monatomic ion** = its **charge**. Alkali metals **+1**, alkaline earth metals **+2**, aluminium **+3**	Na^+ +1, Ba^{2+} +2, Fe^{2+} +2, Al^{3+} +3, I^- −1, O^{2-} −2
3	**Oxygen** = **−2**, except in peroxides = **−1**	MgO, H_2O (−2); H_2O_2, O_2^{2-} (−1)
4	**Hydrogen** = **+1**, except in metal hydrides = **−1**	HCl (+1); LiH, NaH, CaH_2 (−1)
5	**Fluorine** = **−1** always. Other halogens are negative as halide ions but **positive when combined with oxygen** (oxides, oxoacids, oxoanions)	ClF_3: Cl +3; $HClO_4$: Cl +7
6	In a **neutral molecule** the sum is **0**. In a **polyatomic ion** the sum equals the **net charge**	NH_4^+: −3 + 4(+1) = +1

WORKED EXAMPLE · assign oxidation numbers (book Example)

Li_2O: Li is +1 (rule 2), O is −2. Check: 2(+1) + (−2) = 0 ✔

PF_3: F is −1, so P = **+3**

HNO_3: H is +1, O is −2. 1 + x + 3(−2) = 0 → N = **+5**

$Cr_2O_7^{2-}$: 2y + 7(−2) = −2 → y = **+6**

Species	Atom	Oxidation number	Working
MnO_4^-	Mn	+7	x + 4(−2) = −1
H_2S	S	−2	2(+1) + x = 0
SO_2	S	+4	x + 2(−2) = 0
SO_3	S	+6	x + 3(−2) = 0
H_2SO_3	S	+4	2(+1) + x + 3(−2) = 0
NH_4^+	N	−3	x + 4(+1) = +1
H_3AsO_3	As	+3	3(+1) + x + 3(−2) = 0

Species	Atom	Oxidation number	Working
ClF_3	Cl	+3	$x + 3(-1) = 0$

Chlorine oxoacids (book exercise)

Acid	Name in the book	Cl oxidation number
$HClO_4$	Perchloric acid	+7
$HClO_3$	Chloric acid	+5
$HClO_2$	Chlorous acid	+3
HClO	Hypochlorous acid	+1

EXAM HACK

Chlorine ladder: HCl**O** = +1, $HClO_2$ = +3, $HClO_3$ = +5, $HClO_4$ = +7. Every extra oxygen adds **2**. Same pattern for Br and I.

TRAP · examiners love this

Do **not** assume oxygen is always −2 or hydrogen is always +1. Check for **peroxides (O = −1)** and **metal hydrides (H = −1)**. Always confirm the sum equals zero or the net charge.

12.6 Types of redox reactions

Type	What happens	Example
Combustion	Reactions with elemental oxygen are redox	$S + O_2 \rightarrow SO_2$ (S: 0 → +4)
Decomposition	Produces **one or more free elements**	$2HgO \rightarrow 2Hg + O_2$; $2KClO_3 \rightarrow 2KCl + 3O_2$; $2NaH \rightarrow 2Na + H_2$
Displacement	An ion or atom in a compound is replaced by another element: A + BC → AC + B. Three kinds: **hydrogen, metal, halogen**	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
Disproportionation	One element in one oxidation state is **both oxidised and reduced**	$2H_2O_2 \rightarrow 2H_2O + O_2$; $Cl_2 + 2NaOH \rightarrow NaClO + NaCl + H_2O$

Displacement reactions and the activity series

- Use the **activity (electrochemical) series** to predict. **Any metal above hydrogen displaces it from water or acid.** Metals below hydrogen do not react with water or acid.
- In general, a metal **displaces the metals below it** from their compounds. **Zn is above Ag**, so Zn displaces Ag^+ from Ag_2SO_4 solution.
- **Metallurgy:** vanadium is made from V_2O_5 with calcium: $V_2O_5 + 5Ca \rightarrow 2V + 5CaO$. Titanium from $TiCl_4$ with magnesium: $TiCl_4 + 2Mg \rightarrow Ti + 2MgCl_2$. The reducing metal is **above** the metal reduced: Ca above V, Mg above Ti.

Kind	Who displaces what	Example
Hydrogen displacement	**Group 1A and Ca, Sr, Ba** (most reactive) displace H_2 from **cold water**	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
Hydrogen displacement	**Al and Fe** (less reactive) react with **steam**	$2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2$

Kind	Who displaces what	Example
Hydrogen displacement	Some metals displace H_2 from **acids**	$Cd + 2HCl \rightarrow CdCl_2 + H_2$; $Sn + 2HCl \rightarrow SnCl_2 + H_2$
Metal displacement	An uncombined metal displaces a metal ion	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
Halogen displacement	A more reactive halogen displaces a less reactive halide. **$F_2 > Cl_2 > Br_2 > I_2$**	$Cl_2 + 2KBr \rightarrow 2KCl + Br_2$; $Br_2 + 2I^- \rightarrow 2Br^- + I_2$

TRAP · examiners love this

Fluorine is not used for displacement in water because it **reacts vigorously with water**. Remember: **Cl_2 can displace Br^- and I^-**, **Br_2 can displace I^-**, but **I_2 cannot displace Br^- or Cl^-**.

12.7 Resolving a redox reaction into half-reactions

WORKED EXAMPLE · $2Sr + O_2 \rightarrow 2SrO$ (book)

Oxidation numbers: Sr 0 → +2, O 0 → −2.

Oxidation half-reaction: $2Sr \rightarrow 2Sr^{2+} + 4e^-$

Reduction half-reaction: $O_2 + 4e^- \rightarrow 2O^{2-}$

Oxidising agent (oxidant): O_2. Reducing agent (reductant): Sr.

Reaction (book Problem 1)	Oxidation half	Reduction half
$2Ca + O_2 \rightarrow 2CaO$	$2Ca \rightarrow 2Ca^{2+} + 4e^-$	$O_2 + 4e^- \rightarrow 2O^{2-}$
$2Na + S \rightarrow Na_2S$	$2Na \rightarrow 2Na^+ + 2e^-$	$S + 2e^- \rightarrow S^{2-}$
$2Li + H_2 \rightarrow 2LiH$	$2Li \rightarrow 2Li^+ + 2e^-$	$H_2 + 2e^- \rightarrow 2H^-$
$3Mg + N_2 \rightarrow Mg_3N_2$	$3Mg \rightarrow 3Mg^{2+} + 6e^-$	$N_2 + 6e^- \rightarrow 2N^{3-}$
$Zn + I_2 \rightarrow ZnI_2$	$Zn \rightarrow Zn^{2+} + 2e^-$	$I_2 + 2e^- \rightarrow 2I^-$

EXAM HACK

Metal in each reaction is **always the reductant** (it loses electrons). The non-metal (O_2, S, H_2, N_2, I_2) is **always the oxidant** here.

12.8 Balancing redox equations: the ion-electron method

For complex redox reactions involving oxoanions such as chromate (CrO_4^{2-}), dichromate ($Cr_2O_7^{2-}$), permanganate (MnO_4^-), nitrate (NO_3^-) and sulphate (SO_4^{2-}). The overall reaction is split into **two half-reactions**, each is balanced separately, then they are recombined. The book describes **six steps**.

Step	Do this
1	Write the skeletal equation and assign oxidation numbers to every atom
2	Separate into the **reduction** and **oxidation** half-reactions
3	Balance each half-reaction: **atoms** (O with **H_2O**, H with **H^+**) then **charge** with electrons
4	**Equalise the electrons** by multiplying the half-reactions
5	**Add** the two half-reactions and cancel what appears on both sides
6	Write the final equation and **check atoms and charge**

EXAM HACK

Order inside step 3: **atoms other than O and H → O (add H_2O) → H (add H^+) → charge (add e^-)**. In **basic** solution, after adding H^+, add the **same number of OH^- to both sides** and combine H^+ + OH^- into H_2O.

WORKED EXAMPLE · acidic medium (book Example 1): $Cr_2O_7^{2-}$ + C → Cr^{3+} + CO_2 (soil organic carbon test)

Step 1: Cr is +6 in $Cr_2O_7^{2-}$ and +3 in Cr^{3+} (reduced). C is 0 and goes to +4 in CO_2 (oxidised).

Step 2: Reduction: $Cr_2O_7^{2-} \rightarrow Cr^{3+}$. Oxidation: $C \rightarrow CO_2$.

Step 3: Reduction: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$. Oxidation: $C + 2H_2O \rightarrow CO_2 + 4H^+ + 4e^-$.

Step 4: Multiply reduction by **2** and oxidation by **3** so both have **12 e^-**.

$2Cr_2O_7^{2-} + 28H^+ + 12e^- \rightarrow 4Cr^{3+} + 14H_2O$ and $3C + 6H_2O \rightarrow 3CO_2 + 12H^+ + 12e^-$

Step 5: Add and cancel the $12e^-$, $12H^+$ and $6H_2O$.

Step 6: $2Cr_2O_7^{2-} + 16H^+ + 3C \rightarrow 4Cr^{3+} + 8H_2O + 3CO_2$

Check: charge left = 2(−2) + 16 = +12. Charge right = 4(+3) = +12 ✔.

WORKED EXAMPLE · basic medium (book Example 2): MnO_4^- + I^- → MnO_2 + I_2

Step 1: Mn goes +7 → +4 (reduced). I goes −1 → 0 (oxidised).

Step 2: Oxidation: $I^- \rightarrow I_2$. Reduction: $MnO_4^- \rightarrow MnO_2$.

Step 3 (oxidation): $2I^- \rightarrow I_2 + 2e^-$

Step 3 (reduction): $MnO_4^- \rightarrow MnO_2 + 2H_2O$ (add water for O); add $4H^+$ on the left; add $4OH^-$ to **both** sides; combine $4H^+ + 4OH^- = 4H_2O$: **$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$**

Step 4: Oxidation × 3 → $6I^- \rightarrow 3I_2 + 6e^-$. Reduction × 2 → $2MnO_4^- + 4H_2O + 6e^- \rightarrow 2MnO_2 + 8OH^-$.

Steps 5 and 6: $6I^- + 2MnO_4^- + 4H_2O \rightarrow 3I_2 + 2MnO_2 + 8OH^-$

TRAP · examiners love this

Never add H^+ to a **basic** equation in the final answer. If you used it in step 3, convert with OH^- as shown.

Always **check the charge** at the end. Equal atoms with unequal charge means a mistake in step 3.

ALL THE HACKS FOR THIS CHAPTER

OIL RIG (Oxidation Is Loss, Reduction Is Gain) or **LEO says GER**.

The oxidant is the one reduced. The reductant is the one oxidised.

Oxidation numbers: free element 0; O −2 (peroxide −1); H +1 (hydride −1); F −1; total = charge.

Chlorine oxoacids: $HClO$ +1, $HClO_2$ +3, $HClO_3$ +5, $HClO_4$ +7.

Reducing-agent tests: purple $KMnO_4$ → colourless, orange dichromate → green, brown Fe^{3+} → green.

Half-reaction order: atoms → O with H_2O → H with H^+ → charge with e^-. Basic: add OH^- to both sides.

Activity series: a metal displaces anything below it. Halogens: F_2 > Cl_2 > Br_2 > I_2.

QUICK RECALL · read this the night before

Redox = electron transfer. OIL RIG: Oxidation Is Loss, Reduction Is Gain.

Oxidant is reduced (gains e^-). Reductant is oxidised (loses e^-).

Five ways: oxygen, hydrogen, electronegative atoms, electrons, oxidation number.

Tests for reducing agents: $KMnO_4$ purple → colourless; $K_2Cr_2O_7$ orange → green; Fe^{3+} brown → green.

Oxidation numbers: free element 0; O −2 (peroxide −1); H +1 (hydride −1); F −1; sum = charge.

Types: combustion, decomposition, displacement (H, metal, halogen), disproportionation.

Activity series: metals above H displace H from water or acid. Halogens: $F_2 > Cl_2 > Br_2 > I_2$.

Ion-electron method: 6 steps. Acid: H_2O then H^+. Base: add OH^- to both sides.